



אוניברסיטת בן-גוריון בנגב  
Ben-Gurion University of the Negev

## Final project – Malignant Melanoma Survival Analysis

"How do patient characteristics and tumour factors influence melanoma survival?"

Topics in statistics course: 364-2-5091

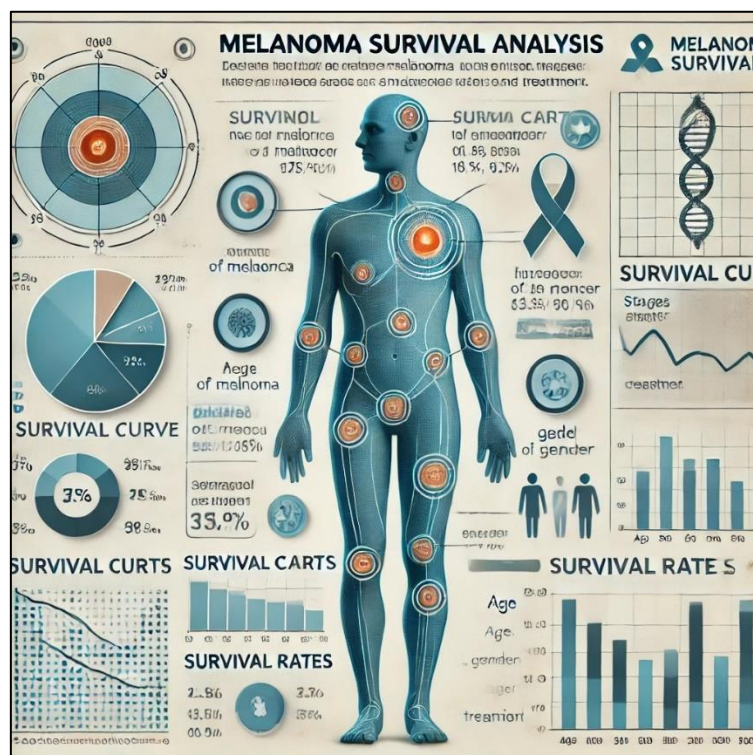
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## **1.Abstract**

This study investigates the influence of patient characteristics and tumor factors on the survival outcomes of individuals diagnosed with malignant melanoma. Using Melanoma survival historical data from 205 patients who underwent surgical treatment at the University Hospital of Odense, Denmark (1962-1977), we performed survival analyses employing Kaplan-Meier curves, log-rank tests, and Cox Proportional Hazards models.

Variables such as age, sex, tumor thickness, ulceration, and their interactions were assessed to determine their impact on survival probabilities.

Results indicate that tumor thickness and ulcer presence are significant predictors of survival, with thicker tumors and ulcerated lesions correlating with decreased survival probability. The selected model is the "Binned Model After Elimination", which is a Cox Proportional Hazard Regression Model (CPH) including the tumor thickness and ulcer variables. The data of each variable was categorized to bins. The model's AIC score is 528.37, and it was selected as the best model according to its AIC score (this model has the lowest AIC score). These findings provide valuable insights into historical melanoma survival factors, highlighting the importance of early detection and tailored interventions.

## **2.Introduction**

Malignant melanoma, which means "black tumour", is the most dangerous type of skin cancer due to its aggressive growth and ability to spread to any organ. It originates from melanocytes, the pigment-producing cells responsible for skin color, and while most melanomas are black or brown, they can also appear pink, red, purple, or skin-colored. Approximately 30% of melanomas develop from existing moles, while the rest arise in normal skin, making it crucial to monitor changes in skin appearance. The number of moles on a person's skin can serve as a predictor for melanoma risk, and understanding one's risk factors is vital for early detection. Melanomas have a 99% cure rate if caught in the earliest stages, but delays in treatment can be life-threatening. Early diagnosis is critical, as treatment success heavily depends on the depth of the cancerous growth.

According to the Global Cancer Observatory (GLOBOCAN), an estimated 325,000 new melanoma cases and 57,000 deaths occurred worldwide in 2020. Men accounted for more cases (174,000) and deaths (32,000) than women (151,000 cases and 25,000 deaths). The highest incidence rates were in Australia and New Zealand, with 42 cases per 100,000 men and 31 per 100,000 women, followed by Western Europe, North America, and Northern Europe. In contrast, melanoma remains rare in most African and Asian countries, with rates below 1 case per 100,000. Mortality rates were highest in New Zealand, at 5 deaths per 100,000. If current trends continue, melanoma cases are projected to rise by 50% to 510,000 cases, with 96,000 deaths (a 68% increase) expected by 2040.

The aim of this work is to examine the relationship between patient characteristics, tumour factors, and survival outcomes in individuals diagnosed with malignant melanoma, specifically focusing on how these factors influence survival probabilities over time. By understanding these relationships, targeted interventions can be developed to improve survival rates and patient outcomes, contributing to advancements in melanoma research.

### 3. Research Goals

The aim of the research is to understand how patient characteristics, like age and sex, and tumor factors, such as thickness and ulceration, affect the survival chances of people with malignant melanoma.

#### 3.1. Research Questions And Hypothesis

**The main research question is:** "How do patient characteristics and tumour factors influence melanoma survival?"

To answer this research question, we will examine six sub-questions and their corresponding hypothesis to provide better analysis:

| Question                                                                                               | Hypothesis                                                                                                                                           |
|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| ➤ What are the survival probabilities of melanoma patients over time?                                  | ➤ <b>No hypothesis</b> (descriptive analysis).                                                                                                       |
| ➤ How do survival probabilities differ between groups?                                                 | ➤ <b>H0:</b> There is no difference in survival probabilities between groups.<br>➤ <b>H1:</b> There is a significant difference.                     |
| ➤ How do patient characteristics and tumour factors influence the hazard rate?                         | ➤ <b>H0:</b> Patient and tumour factors do not significantly influence the hazard rate.<br>➤ <b>H1:</b> At least one factor does.                    |
| ➤ Which predictors are the most significant for melanoma survival?                                     | ➤ <b>H0:</b> None of the predictors significantly influences survival.<br>➤ <b>H1:</b> At least one predictor significantly does.                    |
| ➤ How do the predictors including their interactions influence melanoma survival?                      | ➤ <b>H0:</b> Interaction terms and predictors do not significantly influence survival.<br>➤ <b>H1:</b> They significantly do.                        |
| ➤ Is there a significant difference between the final model (with interactions) and the reduced model? | ➤ <b>H0:</b> The reduced model is as good as the final model with interactions.<br>➤ <b>H1:</b> The final model provides a significantly better fit. |

**Table 1 – Sub questions and their hypothesis.**

This research focuses on understanding the relationships between patient characteristics, tumor factors, and survival outcomes in malignant melanoma. The study will mainly analyze statistical data to explore survival probabilities and identify key prognostic factors.

#### 3.2. Limitations Of The Research

One limitation of this research is that the data is from 1962 to 1977, so it doesn't reflect the survival chances of melanoma patients today, given the progress in medical technology and treatments. This means all recommendations and conclusions are based on what was true during that time. While the results give helpful insights into survival factors back then, they might not fully apply to modern-day care and advances.

## 4.Methods

### 4.1.Data Collection

The data for this research was collected from patients with malignant melanoma who underwent tumour removal surgery at the Department of Plastic Surgery, University Hospital of Odense, Denmark, between the years 1962 and 1977. The dataset consists of 205 rows representing patients who underwent surgery for the complete removal of their tumour, along with approximately 2.5 cm of the surrounding skin. The dataset includes 79 male patients (38.54%) and 126 female patients (61.46%), with 5 children under the age of 18. The youngest patient in the dataset is 4 years old, and the oldest is 95 years old.

### 4.2.Research Variables (Original Data)

| Serial number | Dependent/ Independent variable | Variable name                                                                        | Variable type | Variable description                                                                                                                                                                                        |
|---------------|---------------------------------|--------------------------------------------------------------------------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1             | Not part of the experiment      | patient                                                                              | X             | Patient serial number.                                                                                                                                                                                      |
| 2             | Dependent                       | time                                                                                 | Continuous    | Survival time in days since the operation. Time is censored.                                                                                                                                                |
| 3             | Dependent                       | status<br><b>Note:</b> This will change to <b>event</b> column for survival analysis | Categorical   | 1 - Indicates that they had died from melanoma.<br>2 - Indicates that they were still alive at the end of the data collection.<br>3 - Indicates that they had died from causes unrelated to their melanoma. |
| 4             | Independent                     | sex                                                                                  | Categorical   | 0 – Female.<br>1 – Male.                                                                                                                                                                                    |
| 5             | Independent                     | age                                                                                  | Continuous    | Age in years at the time of the operation.                                                                                                                                                                  |
| 6             | Independent                     | year                                                                                 | Categorical   | Year of operation.                                                                                                                                                                                          |
| 7             | Independent                     | thickness                                                                            | Continuous    | Tumour thickness in mm.                                                                                                                                                                                     |
| 8             | Independent                     | ulcer                                                                                | Categorical   | 1 – Present.<br>0 – absent.                                                                                                                                                                                 |

Table 2 – Research variable explanation.

### 4.3.Data Explanation and Adjustments

#### ➤ Descriptive Statistics:

|       | patient    | time        | status     | sex        | age        | year        | thickness  | ulcer      |
|-------|------------|-------------|------------|------------|------------|-------------|------------|------------|
| count | 205.000000 | 205.000000  | 205.000000 | 205.000000 | 205.000000 | 205.000000  | 205.000000 | 205.000000 |
| mean  | 103.000000 | 2152.800000 | 1.790244   | 0.385366   | 52.463415  | 1969.907317 | 2.919854   | 0.439024   |
| std   | 59.322565  | 1122.060667 | 0.551204   | 0.487873   | 16.671711  | 2.575563    | 2.959433   | 0.497483   |
| min   | 1.000000   | 10.000000   | 1.000000   | 0.000000   | 4.000000   | 1962.000000 | 0.100000   | 0.000000   |
| 25%   | 52.000000  | 1525.000000 | 1.000000   | 0.000000   | 42.000000  | 1968.000000 | 0.970000   | 0.000000   |
| 50%   | 103.000000 | 2005.000000 | 2.000000   | 0.000000   | 54.000000  | 1970.000000 | 1.940000   | 0.000000   |
| 75%   | 154.000000 | 3042.000000 | 2.000000   | 1.000000   | 65.000000  | 1972.000000 | 3.560000   | 1.000000   |
| max   | 205.000000 | 5565.000000 | 3.000000   | 1.000000   | 95.000000  | 1977.000000 | 17.420000  | 1.000000   |

Figure 1 - Descriptive statistics table

This figure focuses mainly on non-categorical variables in the dataset to provide a first impression of the data variables. Here are some key conclusions:

#### ➤ Patient Age:

The age of patients ranges from 4 to 95 years, with a mean age of approximately 52.5 years.

The majority of patients fall between 42 (25<sup>th</sup> percentile) and 65 (75<sup>th</sup> percentile) years of age.

➤ **Time:**

The survival times range from a minimum of 10 to a maximum of 5,565 days, with a median time of 2,005 days.

➤ **Status:**

Most patients (median = 2) are censored data, suggesting they did not die from melanoma during the study.

➤ **Tumor Thickness:**

The distribution is skewed, with 50% of tumors measuring less than 1.94 mm (median), but the presence of very thick tumors (max = 17.42 mm) inflates the mean.

➤ **Year:**

The year of diagnosis spans from 1962 to 1977, with a median year of 1970.

➤ **Data Information and Empty cells:**

|                                       |           |              |       |         |           |   |
|---------------------------------------|-----------|--------------|-------|---------|-----------|---|
| <class 'pandas.core.frame.DataFrame'> |           |              |       |         |           |   |
| RangeIndex: 205 entries, 0 to 204     |           |              |       |         |           |   |
| Data columns (total 8 columns):       |           |              |       |         |           |   |
| #                                     | Column    | Non-Null     | Count | Dtype   |           |   |
| ---                                   | ----      | -----        | ----- | -----   |           |   |
| 0                                     | patient   | 205 non-null |       | int64   | patient   | 0 |
| 1                                     | time      | 205 non-null |       | int64   | time      | 0 |
| 2                                     | status    | 205 non-null |       | int64   | status    | 0 |
| 3                                     | sex       | 205 non-null |       | int64   | sex       | 0 |
| 4                                     | age       | 205 non-null |       | int64   | age       | 0 |
| 5                                     | year      | 205 non-null |       | int64   | year      | 0 |
| 6                                     | thickness | 205 non-null |       | float64 | thickness | 0 |
| 7                                     | ulcer     | 205 non-null |       | int64   | ulcer     | 0 |
| dtypes: float64(1), int64(7)          |           |              |       |         |           |   |
| memory usage: 12.9 KB                 |           |              |       |         |           |   |

Figure 2 - On the left we can see data types and on the right, we checked for empty cells in the data.

This phase helps us identify any missing cells in the data, which would require handling, such as deleting rows or filling cells with calculated values. As we can see, there are no empty cells, and all columns contain the same number of entries.

➤ **Event Column:**

The original status column indicated whether patients had died from melanoma (1), were still alive (2), or had died from causes unrelated to melanoma (3). To simplify and align the data for survival analysis, it was transformed into a binary event column. In the new column, "1" represents death specifically due to melanoma, while "2" (alive) and "3" (death from other causes) were grouped as "0" to indicate censored data. This transformation allows the analysis to focus on survival outcomes

related to melanoma. Censoring, is a main aspect of survival data, is effectively addressed with this column. Without it, survival models would face challenges in handling incomplete observations, potentially leading to biased or invalid results.

|   | patient | time | status | sex | age | year | thickness | ulcer | event |
|---|---------|------|--------|-----|-----|------|-----------|-------|-------|
| 0 | 1       | 10   | 3      | 1   | 76  | 1972 | 6.76      | 1     | 0     |
| 1 | 2       | 30   | 3      | 1   | 56  | 1968 | 0.65      | 0     | 0     |
| 2 | 3       | 35   | 2      | 1   | 41  | 1977 | 1.34      | 0     | 0     |
| 3 | 4       | 99   | 3      | 0   | 71  | 1968 | 2.90      | 0     | 0     |
| 4 | 5       | 185  | 1      | 1   | 52  | 1965 | 12.08     | 1     | 1     |

Figure 3 - New event column added to the original data.

**Note:** in this study we have a 2D dependent variable which consists of time and event.

### ➤ Continuous Data Analysis:

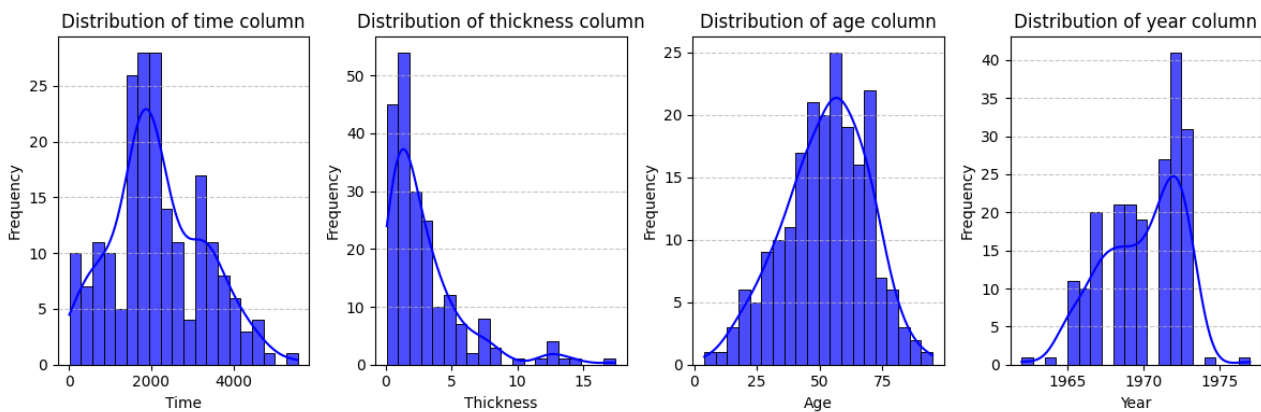


Figure 4 - Continuous data distribution with histogram plot.

These graphs were created to visualize the distributions of continuous variables in the dataset, providing understanding of their patterns and identifying skews that might influence further analysis:

- **Time:**

The survival time distribution appears positively skewed, with most observations clustering around 2000 days. A few cases have significantly longer survival times, creating a long tail.

- **Tumor Thickness:**

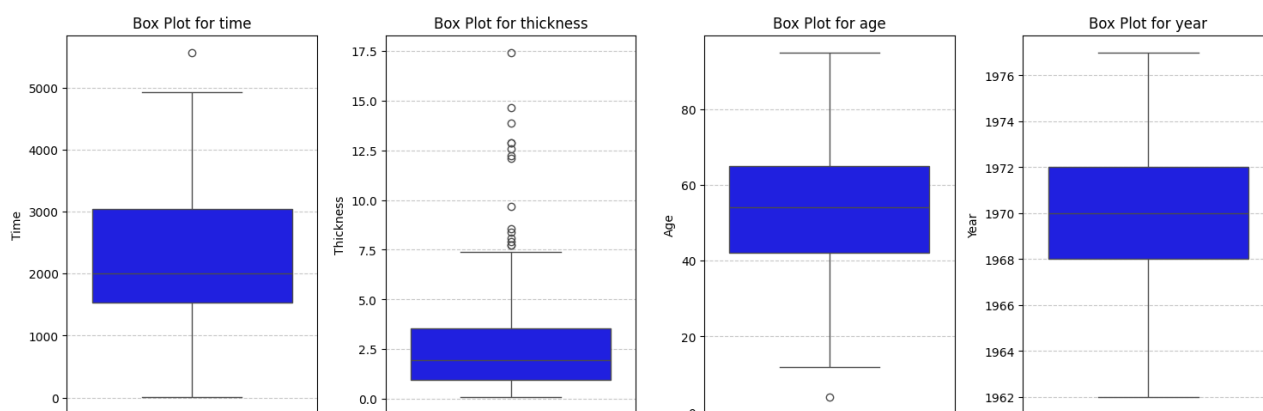
Tumor thickness shows a right-skewed distribution, indicating that thinner tumors are more common, with a few cases having significantly higher thickness.

- **Age:**

The age distribution is close to normal distribution, with a peak around 50-60 years. There is a reasonable spread across younger and older ages.

- **Year:**

The distribution of the year is slightly right skewed, indicating that more patients underwent operations during the later years of the study period (1970 -1977).



**Figure 5 - Box plot for detecting outlier in a continues column.**

The box plots were created to visually summarize the distribution, central tendency, and spread of the main numerical variables in the dataset and identifying outliers:

- **Time:**

A few high outliers are observed, representing exceptionally long survival durations.

- **Tumor Thickness:**

There are several high outliers, indicating some patients with significantly thicker tumors.

- **Age:**

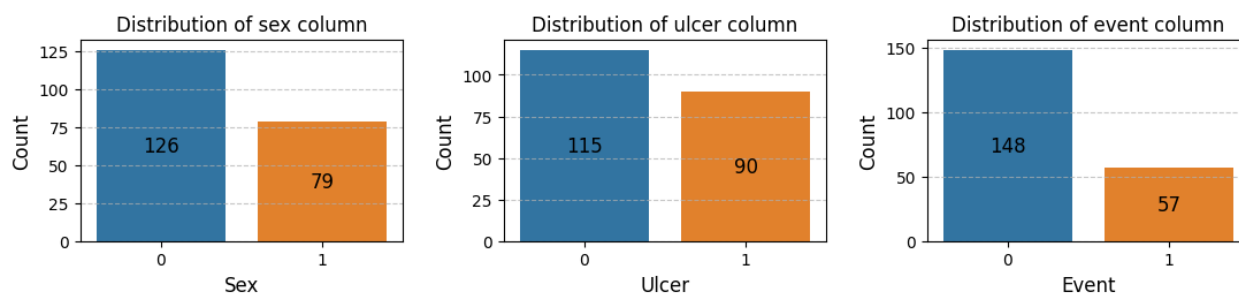
A few outliers at the lower end represent younger patients, possibly pediatric cases.

- **Year:**

The distribution shows no outliers.

In medical research, outliers often correspond to rare cases that can provide valuable insights into survival patterns. Removing these outliers' risks losing critical information that could help understand melanoma survival and inform clinical decisions. Therefore, all data points, stayed the same as it was.

➤ **Categorical Data Analysis:**



**Figure 6 - Categorical data distribution and balance.**

The bar plots help us understand the distribution of categorical variables and assess the balance between their categories:

- **Sex:**

The dataset contains more females than males, but the difference is not very high, so the balance between is close to good for analysis.

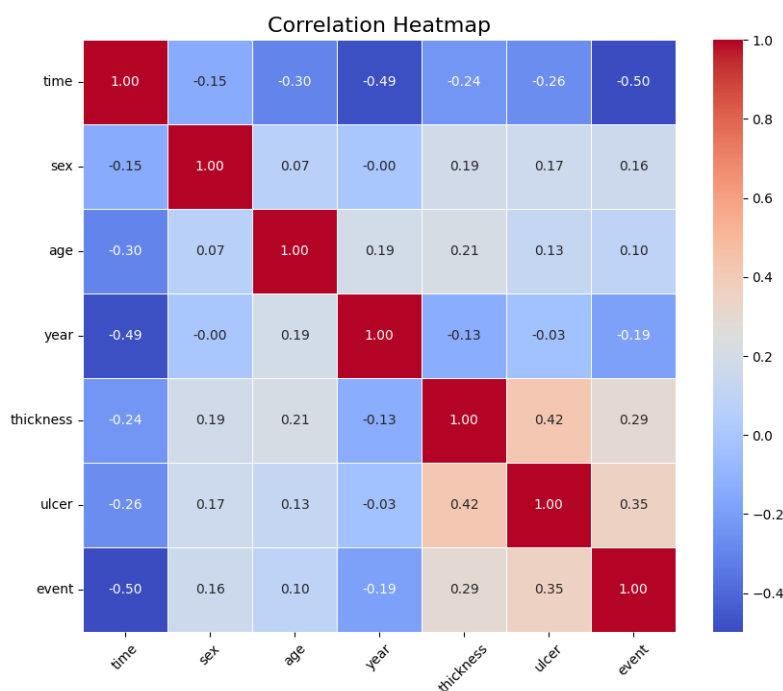
- **Ulcer:**

There is a slight imbalance, with more patients having no ulcer than having an ulcer. This difference is manageable.

- **Event:**

There is a significant imbalance, with more censored events (no death from melanoma) compared to the uncensored events (death from melanoma). This could affect the power of the survival analysis and may cause overfit but because of limitations of it being medical data we will leave it as it is.

➤ **Correlation map:**



**Figure 7 - Correlation map.**

The heatmap shows how the variables in the dataset are related to each other to check for multicollinearity, which happens when two or more variables are too closely related. From the map, we can see that none of the variables are strongly related, as most of the correlations are much lower than the level (around 0.7 or higher) where problems usually happen.

#### 4.4.Data Preprocessing (Before Statistic Tests)

➤ **Data Binning**

As part of the preprocessing phase, continuous variables such as tumour thickness, age, and year of diagnosis were converted into categorical bins to examine differences between groups in survival models using appropriate methods. This process also helps analyse which predictors show differences before fitting the

model, potentially allowing us to drop columns that do not contribute significantly when we separate them to bins. Here we will present an explanation of the binning phase for each variable:

- **Tumour Thickness:**

Tumour thickness was divided into clinically meaningful categories using thresholds guided by research evidence from *Baade et al. (2020)* in the International Journal of Cancer. This study showed that melanoma survival outcomes vary significantly based on thickness. The bins created were:

|                                                             |
|-------------------------------------------------------------|
| ❖ 0-0.8 mm: Thin tumours with lower mortality risk.         |
| ❖ 0.8-1.0 mm: Slightly thicker tumours with increased risk. |
| ❖ 1.0-2.0 mm                                                |
| ❖ 2.0-3.0 mm                                                |
| ❖ 3.0-4.0 mm                                                |
| ❖ 4.0-5.0 mm                                                |
| ❖ 5.0-6.0 mm                                                |
| ❖ >6.0 mm:                                                  |

These categories help reflect increasing tumour severity, as thicker melanomas have a higher risk of mortality.

- **Age:**

Age was grouped into standard intervals to analyze survival trends across clinical age groups and was calculated like this:  $\frac{Max-Min}{number\ of\ bins} = \frac{95\ (oldest)-4\ (youngest)}{5} = 18$  so the bins we get are:

|                                            |
|--------------------------------------------|
| ❖ 0-17 years: Young children and teenagers |
| ❖ 18-34 years: Young adults                |
| ❖ 35-54 years: Middle-aged adults.         |
| ❖ 55-74 years: Older adults                |
| ❖ 75+ years: Elderly patients              |

These intervals provide practical insight into how age affects melanoma outcomes.

- **Year:**

The year column was divided into three intervals and was calculated like this:  $\frac{Max-Min}{number\ of\ bins} = \frac{1977-1962}{3} = 5$  so the bins we get are:

|             |
|-------------|
| ❖ 1962-1967 |
| ❖ 1967-1972 |
| ❖ 1972-1977 |

These categories reflect changes over time, allowing us to analyze potential trends in survival outcomes as medical technology and treatments may advance during this period.

The data consist of bins and the original columns and now looks like this:

| patient | time | status | sex | age | year | thickness | ulcer | event | tumor_thickness_category_mm | age_category | year_category |           |
|---------|------|--------|-----|-----|------|-----------|-------|-------|-----------------------------|--------------|---------------|-----------|
| 0       | 1    | 10     | 3   | 1   | 76   | 1972      | 6.76  | 1     | 0                           | >6.0         | 75+           | 1967-1972 |
| 1       | 2    | 30     | 3   | 1   | 56   | 1968      | 0.65  | 0     | 0                           | 0-0.8        | 55-74         | 1967-1972 |
| 2       | 3    | 35     | 2   | 1   | 41   | 1977      | 1.34  | 0     | 0                           | 1.0-2.0      | 35-54         | 1972-1977 |
| 3       | 4    | 99     | 3   | 0   | 71   | 1968      | 2.90  | 0     | 0                           | 2.0-3.0      | 55-74         | 1967-1972 |
| 4       | 5    | 185    | 1   | 1   | 52   | 1965      | 12.08 | 1     | 1                           | >6.0         | 35-54         | 1962-1967 |

Figure 8 – The data after creating bins.

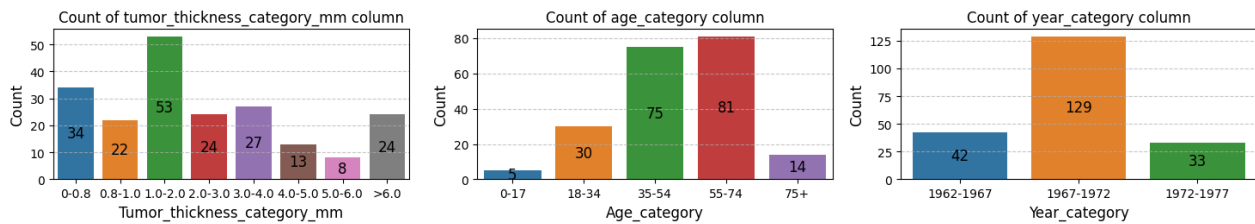


Figure 9 - Binned Data distribution.

## 4.5.Models We Chose And Their Assumptions

### ➤ Statistical methods and tests

| Tests                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Explanation                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>➤ <b>Question 1</b></p> <p>❖ <b>Kaplan-Meier Curve:</b> Using time and event columns to estimate overall survival probabilities for all patients.</p> <p>❖ <b>Visualizing Survival Trends:</b> Creating a single Kaplan-Meier curve to observe survival probabilities over time.</p>                                                                                                                                                                                                                         | <p>➤ Provides an overview of how melanoma patients' survival probabilities change over time, this part is the foundation for further analysis with the data.</p>             |
| <p>➤ <b>Question 2</b></p> <p>❖ <b>Kaplan-Meier Curves for Groups:</b> We plotted survival curves for groups (e.g., males vs. females).</p> <p>❖ <b>Multivariate Log-rank Test:</b> Testing for statistically significant differences in survival distributions across groups.</p> <p>❖ <b>Pair wise Log-Rank Test:</b> Testing in groups differences.</p>                                                                                                                                                      | <p>➤ With this question we explored whether survival probabilities are different between specific groups, identifying trends and testing their statistical significance.</p> |
| <p>➤ <b>Question 3</b></p> <p>❖ <b>Cox Proportional Hazards (CPH) Model:</b> Includes patient characteristics and tumour factors as covariates. On continues variable model and categorized variable model.</p> <p>❖ <b>Interpret Hazard Ratios (HRs):</b> Identified whether factors increase (<math>HR &gt; 1</math>) or decrease (<math>HR &lt; 1</math>) the likelihood of earlier survival events.</p> <p>❖ <b>Validate Model:</b> Checked for proportional hazards assumption (Schoenfeld residuals).</p> | <p>➤ Here we investigated how patient and tumour factors influence survival timing, identifying significant predictors without treating time as the dependent variable.</p>  |
| <p>➤ <b>Question 4</b></p> <p>❖ <b>Stepwise Model Selection:</b> Using backward selection to refine variables on chosen model from question 3.</p>                                                                                                                                                                                                                                                                                                                                                              | <p>➤ This question main task is to analyse the most influential factors we can evaluate what predictors are significant for the model predictions.</p>                       |

|                                                                                                                                                                                                                                                                                                                                               |                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| ❖ Run the model again after stepwise elimination and intercept it like in question 3.                                                                                                                                                                                                                                                         |                                                                                                 |
| ➤ <b>Question 5</b><br>❖ <b>Cox Proportional Hazards (CPH) Model:</b> With a full interaction model.<br>❖ <b>Interpret Hazard Ratios (HRs):</b> Identified whether factors increase ( $HR > 1$ ) or decrease ( $HR < 1$ ) the likelihood of earlier survival events.<br>❖ <b>Validate Model:</b> Checked for proportional hazards assumption. | ➤ Here we investigated how patient and tumour factors and interaction influence survival timing |
| ➤ <b>Question 6</b><br>❖ <b>Likelihood Ratio Test:</b> to compare between the models using log-likelihood and chi square.                                                                                                                                                                                                                     | ➤ Helps to investigate whether the model did or did not improve after interaction terms.        |

**Table 3 - Statical tests and their explanation.**

### ➤ Assumptions for statistical methods

| Test                                   | Assumptions                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ➤ Kaplan-Meier Curve                   | ➤ Censored patients have the same likelihood of survival as those continuing in the study.<br>➤ Survival probabilities are the same whether individuals enter a study early or late.<br>➤ Events (e.g., deaths) occur independently among patients.                                                                                                                                        |
| ➤ Log-Rank Test                        | ➤ The Log-Rank test shares the same assumptions as the Kaplan-Meier survival curve. These include censoring is unrelated to prognosis, survival probabilities are consistent for individuals recruited at different times during the study, and events occurred at the recorded times.                                                                                                     |
| ➤ Cox Proportional Hazards (CPH) Model | ➤ The hazard ratio between groups is assumed to remain constant over time.<br>➤ Survival times of individuals are assumed to be independent of one another.<br>➤ Censored individuals are assumed to have the same risk as those who continue to be followed.<br>➤ The CPH model assumes a log-linear relationship between the predictors and the hazard ratio (We will check if it does). |
| Conditions                             |                                                                                                                                                                                                                                                                                                                                                                                            |
| ➤ CPH model                            | ➤ Picked by AIC criteria                                                                                                                                                                                                                                                                                                                                                                   |
| ➤ Stepwise Model Selection             | ➤ Variables are selected based on P-Value.                                                                                                                                                                                                                                                                                                                                                 |

**Table 4 - Assumptions and conditions for statical tests.**

## 5.Results

### 5.1.Question 1

We used Kaplan-Meier Curve Using time and event columns to estimate overall survival probabilities for all patients. According to the Kaplan Meir survival curve, as the time from the surgery progress the survival chances of the patient decreases. This makes sense, because patients face cumulative risks such as disease progression, treatment complications, aging, and comorbidities, all of which increase the likelihood of adverse events as time progresses.

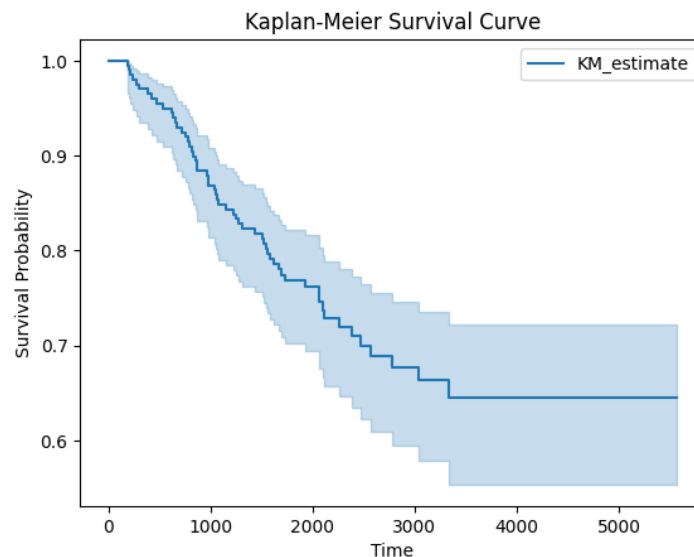


Figure 10 - Kaplan Meir survival curve.

### 5.2.Question 2

#### ➤ Kaplan-Meier Curves for Groups:

The Kaplan-Meier survival curves were created for each category to visualize the survival distributions across the different groups within these variables. This method provides a graphical representation of how survival probabilities change over time, enabling a clearer understanding of whether and how survival differs between groups.

- **Kaplan-Meier Survival Curves for age category (Age after binning):**

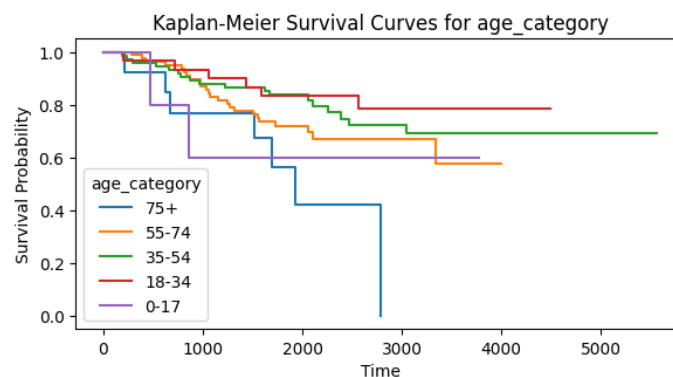
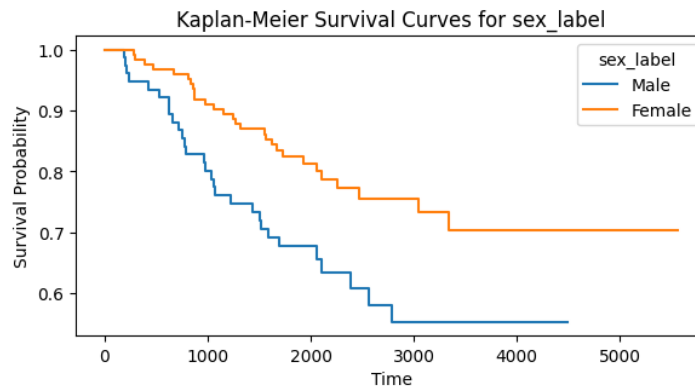


Figure 11 - Kaplan Meir survival curve for age groups.

This graph shows survival curves for different age categories (0-17, 18-34, 35-54, 55-74, 75+). The oldest and youngest age groups (0-17, 75+) demonstrate lower survival probabilities compared to other age groups. This pattern could suggest that individuals in this group may be experiencing more aggressive or specific types of conditions affecting survival. In contrast, middle-aged groups like 18-34 and 35-54 show relatively higher survival probabilities, while older groups (55-74, 75+) experience expected declines due to age-related vulnerabilities. The 0-17 pattern deviates from general expectations and may indicate unique challenges for this demographic.

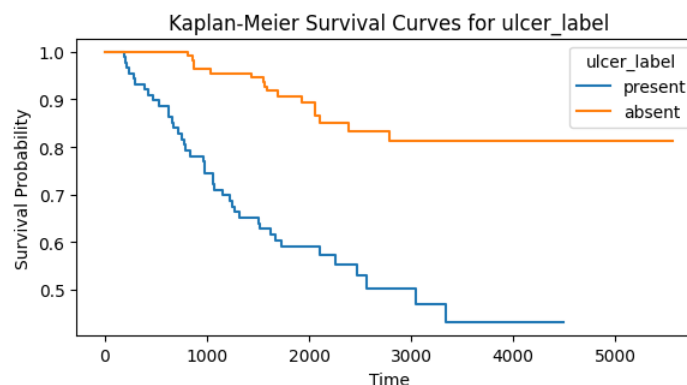
- **Kaplan-Meier Survival Curves for sex (Already binary):**



**Figure 12 - Kaplan Meir survival curve for sex groups (Males and Females).**

This graph illustrates survival curves based on sex (Male and Female). Females show higher survival probabilities, with their curve remaining above the male curve. This aligns with known biological differences and healthcare-seeking behaviors, where females often have better survival outcomes in certain conditions (Stidham et al., n.d.).

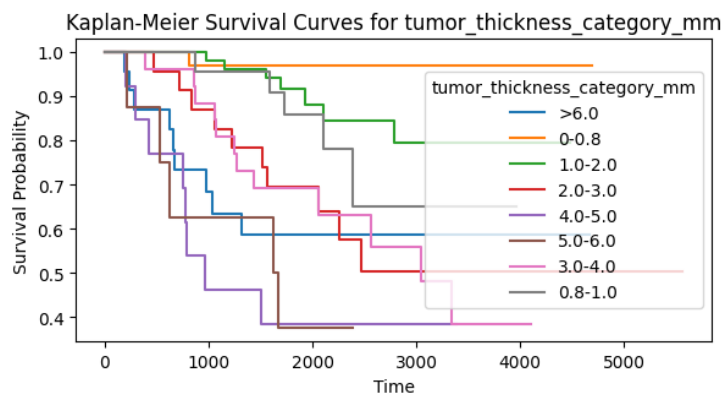
- **Kaplan-Meier Survival Curves for ulcer (Already binary):**



**Figure 13 - Kaplan Meir survival curve for ulcer (present or absent).**

This graph compares survival probabilities based on the presence or absence of an ulcer. Patients without ulcers (absent) have significantly higher survival probabilities than those with ulcers (present). This is logical, as the presence of an ulcer may indicate more aggressive or advanced disease, contributing to worse survival outcomes.

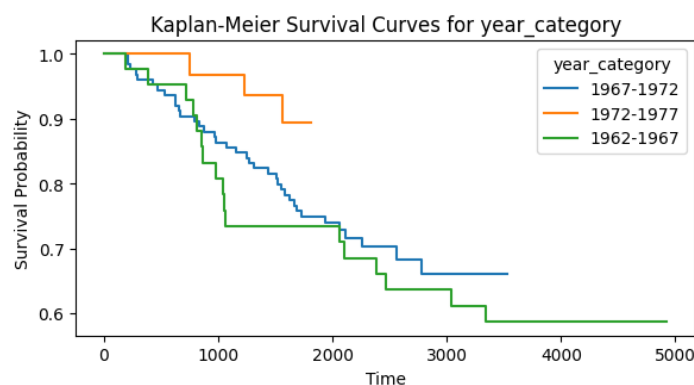
- **Kaplan-Meier Survival Curves for tumor thickness category ( thickness after binning):**



**Figure 14 - Kaplan Meir survival curve for thickness groups.**

This graph presents survival curves for different tumor thickness categories (e.g., 0.8-1.0 mm, 1.0-2.0 mm, etc.). In general, as tumor thickness increases, survival probabilities decrease. From tumor thickness of 5 mm and above the survival probabilities decline rapidly. This pattern makes sense because thicker tumors are associated with more advanced disease stages and worse prognosis.

- **Kaplan-Meier Survival Curves for year category (year after binning):**



**Figure 15 - Kaplan Meir survival curve for year groups.**

This graph shows survival probabilities across different time periods (1962-1967, 1967-1972, 1972-1977). Later periods (1972-1977) exhibit better survival probabilities compared to earlier periods (1962-1967). This trend is expected, as medical treatments, early detection, and healthcare access improved over time, leading to better survival outcomes.

➤ **Multivariate Log-rank Test:**

We performed a log-rank test for evaluation whether there are significant differences in survival distributions between groups in each of the explaining variables. The null hypothesis in each variable is that there is no significant difference between groups in each group.

- For age category variable the Log-Rank test results are:

Global Log-Rank Test for age\_category:  
Test statistic: 10.794237195776004  
p-value: 0.02897647665505408  
Significant differences exist between groups in age\_category.

|                    |                           |          |      |
|--------------------|---------------------------|----------|------|
| t_0                | -1                        |          |      |
| null_distribution  | chi squared               |          |      |
| degrees_of_freedom | 4                         |          |      |
| test_name          | multivariate_logrank_test |          |      |
| test_statistic     | p                         | -log2(p) |      |
| 0                  | 10.79                     | 0.03     | 5.11 |

Figure 16 - age category log rank test results table.

The global log-rank test yielded a test statistic of 10.79 and a p-value of 0.0289, which is below the significance threshold of 0.05. This indicates that significant differences exist in survival outcomes among the groups in the age category variable, leading to the rejection of the null hypothesis.

- For sex variable the Log-Rank test results are:

Global Log-Rank Test for sex:

Test statistic: 6.467976857341599

p-value: 0.010983549936573913

Significant differences exist between groups in sex.

|                    |                           |          |      |
|--------------------|---------------------------|----------|------|
| t_0                | -1                        |          |      |
| null_distribution  | chi squared               |          |      |
| degrees_of_freedom | 1                         |          |      |
| test_name          | multivariate_logrank_test |          |      |
| test_statistic     | p                         | -log2(p) |      |
| 0                  | 6.47                      | 0.01     | 6.51 |

Figure 17 - sex log rank test results table.

The global log-rank test produced a test statistic of 6.47 and a p-value of 0.011, which is below the 0.05 significance threshold. This indicates significant differences in survival outcomes between the groups in the sex variable, leading to the rejection of the null hypothesis.

- For year category variable the Log-Rank test results are:

```
Global Log-Rank Test for year_category:
Test statistic: 3.439379941220874
p-value: 0.32870986372587024
No significant differences detected for year_category.
```

|                    |                           |
|--------------------|---------------------------|
| t_0                | -1                        |
| null_distribution  | chi squared               |
| degrees_of_freedom | 3                         |
| test_name          | multivariate_logrank_test |
| test_statistic     | p -log2(p)                |
| 0                  | 3.44 0.33 1.61            |

Figure 18 - year category log rank test results table.

The log-rank test for the year category variable yielded a test statistic of 3.44 and a p-value of 0.33, which is above the 0.05 threshold. This indicates no significant differences in survival distributions between groups in the year category variable. As a result, this variable was **deleted** from the model because the lack of significant differences suggests it does not provide meaningful information for predicting the outcome. Including variables with weak predictive power can introduce noise, reduce model interpretability, and potentially affect its overall performance of the model.

- For tumour thickness category variable the Log-Rank test results are:

```
Global Log-Rank Test for tumor_thickness_category_mm:
Test statistic: 43.22376396924915
p-value: 3.0196529197843964e-07
Significant differences exist between groups in tumor_thickness_category_mm.
```

|                    |                           |
|--------------------|---------------------------|
| t_0                | -1                        |
| null_distribution  | chi squared               |
| degrees_of_freedom | 7                         |
| test_name          | multivariate_logrank_test |
| test_statistic     | p -log2(p)                |
| 0                  | 43.22 <0.005 21.66        |

Figure 19 - thickness category log rank test results table.

The global log-rank test produced a test statistic of 43.22 and a p-value smaller than 0.005, which is below the 0.05 significance threshold. This indicates significant differences in survival outcomes between the groups in the sex tumour thickness, leading to the rejection of the null hypothesis.

- For ulcer variable the Log-Rank test results are:

Global Log-Rank Test for ulcer:  
 Test statistic: 29.56298542809486  
 p-value: 5.412879724223082e-08  
 Significant differences exist between groups in ulcer.

|                    |                           |
|--------------------|---------------------------|
| t_0                | -1                        |
| null_distribution  | chi squared               |
| degrees_of_freedom | 1                         |
| test_name          | multivariate_logrank_test |
| test_statistic     | p -log2(p)                |
| 0                  | 29.56 <0.005 24.14        |

Figure 20 - ulcer log rank test results table.

The global log-rank test produced a test statistic of 29.56 and a p-value smaller than 0.005, which is below the 0.05 significance threshold. This indicates significant differences in survival outcomes between the groups in the sex ulcer, leading to the rejection of the null hypothesis.

#### ➤ Pairwise Log-Rank Tests for inner groups differences

We ran pairwise log-rank tests for the age and thickness categories, as they have multiple groups, unlike binary columns like sex and ulcer where differences are already clear. The goal was to see if any groups within these categories had similar survival patterns. If no significant differences were found, we merged the groups to simplify the data and make the model easier to understand. This keeps the model focused on important differences.

#### ➤ Age category variable Pairwise Log-Rank test results

|                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------|
| ❖ 75+ vs 55-74: p-value = 0.0479. The null hypothesis is rejected, indicating a significant difference in survival patterns between these groups. |
| ❖ 75+ vs 35-54: p-value = 0.0021. The null hypothesis is rejected, showing a significant difference between these groups.                         |
| ❖ 75+ vs 18-34: p-value = 0.0044. The null hypothesis is rejected, suggesting a significant difference between these groups.                      |
| ❖ 75+ vs 0-17: p-value = 0.4635. The null hypothesis is not rejected, indicating no significant difference between these groups.                  |
| ❖ 55-74 vs 35-54: p-value = 0.2155. The null hypothesis is not rejected, showing no significant difference between these groups.                  |
| ❖ 55-74 vs 18-34: p-value = 0.1484. The null hypothesis is not rejected, indicating no significant difference between these groups.               |
| ❖ 55-74 vs 0-17: p-value = 0.6780. The null hypothesis is not rejected, showing no significant difference between these groups.                   |
| ❖ 35-54 vs 18-34: p-value = 0.5100. The null hypothesis is not rejected, suggesting no significant difference between these groups.               |
| ❖ 35-54 vs 0-17: p-value = 0.3871. The null hypothesis is not rejected, indicating no significant difference between these groups.                |
| ❖ 18-34 vs 0-17: p-value = 0.2437. The null hypothesis is not rejected, showing no significant difference between these groups.                   |

Based on these results, we merged the age categories as follows (see [Appendix 8.1](#)):

- ❖ The groups **18-34**, **35-54**, and **55-74** were merged into another single category (1).
- ❖ The groups **0-17** and **75+** were merged into a single category (2).

➤ **Thickness category variable Pairwise Log-Rank test results**

|                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------|
| ❖ <b>&gt;6.0 vs 0-0.8</b> : p-value = 0.0003. The null hypothesis is rejected, indicating a significant difference in survival patterns.        |
| ❖ <b>&gt;6.0 vs 1.0-2.0</b> : p-value = 0.0039. The null hypothesis is rejected, indicating a significant difference in survival patterns.      |
| ❖ <b>&gt;6.0 vs 2.0-3.0</b> : p-value = 0.7538. The null hypothesis is not rejected, indicating no significant difference in survival patterns. |
| ❖ <b>&gt;6.0 vs 4.0-5.0</b> : p-value = 0.2795. The null hypothesis is not rejected, indicating no significant difference in survival patterns. |
| ❖ <b>&gt;6.0 vs 5.0-6.0</b> : p-value = 0.3876. The null hypothesis is not rejected, indicating no significant difference in survival patterns. |
| ❖ <b>&gt;6.0 vs 3.0-4.0</b> : p-value = 0.9180. The null hypothesis is not rejected, indicating no significant difference in survival patterns. |
| ❖ <b>&gt;6.0 vs 0.8-1.0</b> : p-value = 0.1289. The null hypothesis is not rejected, indicating no significant difference in survival patterns. |
| ❖ <b>0-0.8 vs 1.0-2.0</b> : p-value = 0.0965. The null hypothesis is not rejected, indicating no significant difference in survival patterns.   |
| ❖ <b>0-0.8 vs 2.0-3.0</b> : p-value = 0.0003. The null hypothesis is rejected, indicating a significant difference in survival patterns.        |
| ❖ <b>0-0.8 vs 4.0-5.0</b> : p-value = 0.0000. The null hypothesis is rejected, indicating a significant difference in survival patterns.        |
| ❖ <b>0-0.8 vs 5.0-6.0</b> : p-value = 0.0000. The null hypothesis is rejected, indicating a significant difference in survival patterns.        |
| ❖ <b>0-0.8 vs 3.0-4.0</b> : p-value = 0.0001. The null hypothesis is rejected, indicating a significant difference in survival patterns.        |
| ❖ <b>0-0.8 vs 0.8-1.0</b> : p-value = 0.0207. The null hypothesis is rejected, indicating a significant difference in survival patterns.        |
| ❖ <b>1.0-2.0 vs 2.0-3.0</b> : p-value = 0.0070. The null hypothesis is rejected, indicating a significant difference in survival patterns.      |
| ❖ <b>1.0-2.0 vs 4.0-5.0</b> : p-value = 0.0000. The null hypothesis is rejected, indicating a significant difference in survival patterns.      |
| ❖ <b>1.0-2.0 vs 5.0-6.0</b> : p-value = 0.0001. The null hypothesis is rejected, indicating a significant difference in survival patterns.      |
| ❖ <b>1.0-2.0 vs 3.0-4.0</b> : p-value = 0.0023. The null hypothesis is rejected, indicating a significant difference in survival patterns.      |
| ❖ <b>1.0-2.0 vs 0.8-1.0</b> : p-value = 0.3409. The null hypothesis is not rejected, indicating no significant difference in survival patterns. |
| ❖ <b>2.0-3.0 vs 4.0-5.0</b> : p-value = 0.0919. The null hypothesis is not rejected, indicating no significant difference in survival patterns. |
| ❖ <b>2.0-3.0 vs 5.0-6.0</b> : p-value = 0.1659. The null hypothesis is not rejected, indicating no significant difference in survival patterns. |
| ❖ <b>2.0-3.0 vs 3.0-4.0</b> : p-value = 0.8424. The null hypothesis is not rejected, indicating no significant difference in survival patterns. |
| ❖ <b>2.0-3.0 vs 0.8-1.0</b> : p-value = 0.1949. The null hypothesis is not rejected, indicating no significant difference in survival patterns. |
| ❖ <b>4.0-5.0 vs 5.0-6.0</b> : p-value = 0.9024. The null hypothesis is not rejected, indicating no significant difference in survival patterns. |
| ❖ <b>4.0-5.0 vs 3.0-4.0</b> : p-value = 0.0773. The null hypothesis is not rejected, indicating no significant difference in survival patterns. |

|                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------|
| ❖ <b>4.0-5.0 vs 0.8-1.0:</b> p-value = 0.0048. The null hypothesis is rejected, indicating a significant difference in survival patterns.      |
| ❖ <b>5.0-6.0 vs 3.0-4.0:</b> p-value = 0.1342. The null hypothesis is not rejected, indicating no significant difference in survival patterns. |
| ❖ <b>5.0-6.0 vs 0.8-1.0:</b> p-value = 0.0087. The null hypothesis is rejected, indicating a significant difference in survival patterns.      |
| ❖ <b>3.0-4.0 vs 0.8-1.0:</b> p-value = 0.1849. The null hypothesis is not rejected, indicating no significant difference in survival patterns. |

Based on these results, we merged the thickness categories as follows (see [Appendix 8.2](#)):

- ❖ The groups **0-0.8, 0.8-1.0, and 1.0-2.0** were merged into a single category (1).
- ❖ The groups **2.0-3.0, 3.0-4.0, 4.0-5.0, 5.0-6.0, and >6.0** were merged into a single category (2).

After merging the groups, we created dummy variables for the categories to convert them into a numerical format required by most models. This allows the model to process categorical data. Creating dummy variables ensures the model captures the differences between the merged tumour thickness groups and age groups.

**The data now looks like this:**

| patient | time | status | sex | age | year | thickness | ulcer | event | tumor_thickness_category_mm | age_category | year_category | sex_label | ulcer_label | age_category_new_1 | tumor_thickness_category_mm_new_2 |       |
|---------|------|--------|-----|-----|------|-----------|-------|-------|-----------------------------|--------------|---------------|-----------|-------------|--------------------|-----------------------------------|-------|
| 0       | 1    | 10     | 3   | 1   | 76   | 1972      | 6.76  | 1     | 0                           | >6.0         | 75+           | 1967-1972 | Male        | present            | False                             | True  |
| 1       | 2    | 30     | 3   | 1   | 56   | 1968      | 0.65  | 0     | 0                           | 0-0.8        | 55-74         | 1967-1972 | Male        | absent             | True                              | False |
| 2       | 3    | 35     | 2   | 1   | 41   | 1977      | 1.34  | 0     | 0                           | 1.0-2.0      | 35-54         | 1972-1977 | Male        | absent             | True                              | False |
| 3       | 4    | 99     | 3   | 0   | 71   | 1968      | 2.90  | 0     | 0                           | 2.0-3.0      | 55-74         | 1967-1972 | Female      | absent             | True                              | True  |
| 4       | 5    | 185    | 1   | 1   | 52   | 1965      | 12.08 | 1     | 1                           | >6.0         | 35-54         | 1962-1967 | Male        | present            | True                              | True  |

**Figure 21 - data after all manipulations.**

**Note:** in the next question we will explain what variables were deleted as they have no need. For example: the labels we used for the graphs and the categories before manipulations.

### 5.3.Question 3

In this question, we built two data structures for CPH model to compare how data representation affects the results. The binned model grouped continuous variables, like age and thickness, into categories for easier interpretation, while the continuous model kept these variables in their original form to capture real relationships. We selected the best model based on AIC values and then performed backward stepwise elimination to remove insignificant variables for an optimized model. In the end, we ran the model again to see if it had improved. Before running the models, we removed irrelevant columns and checked for multicollinearity using Variance Inflation Factor (VIF). If VIF values were smaller than 5, it will indicate that multicollinearity is not an issue and that the variables that we will use in the model are not highly correlated.

#### ➤ Columns Deleted:

**status:** not relevant, we created a combined column called event as stated before.

**tumor\_thickness\_category\_mm:** not relevant, we created categorized dummy variable.

**age\_category:** not relevant, we created categorized dummy variable.

**year\_category:** no difference in groups, we deleted it for potentially being a weak predictor.

**sex\_label:** was used only for graph legend.

**ucler\_label:** was used only for graph legend.

➤ **VIF For Multicollinearity And Variables For Each Model:**

| Binned Model |                                   |          | Continuous Model |           |          |
|--------------|-----------------------------------|----------|------------------|-----------|----------|
|              | Variable                          | VIF      |                  | Variable  | VIF      |
| 0            | time                              | 1.374726 | 0                | time      | 1.472698 |
| 1            | sex                               | 1.068202 | 1                | sex       | 1.057871 |
| 2            | ulcer                             | 1.430158 | 2                | age       | 1.137503 |
| 3            | event                             | 1.520394 | 3                | thickness | 1.308527 |
| 4            | age_category_new_1                | 1.050996 | 4                | ulcer     | 1.323774 |
| 5            | tumor_thickness_category_mm_new_2 | 1.469929 | 5                | event     | 1.468263 |

Table 5 – VIF and variables for each model

In both cases we can see all variables are lower than 5 which means there is no multicollinearity and that the variables are not highly correlated.

➤ **CPH Model**

$$h(t|X) = h_0(t) \cdot e^{\beta_1 \cdot X_1 + \beta_2 \cdot X_2 + \dots + \beta_p \cdot X_p}$$

Where:

- $h(t|X)$ : The hazard function at time  $t$ , given predictor variables  $X$ .
- $h_0(t)$ : The baseline hazard function (when all predictors are zero).
- $\beta_1, \beta_2, \dots, \beta_p$ : The regression coefficients associated with the predictor variables  $X_1, X_2, \dots, X_p$
- $X_1, X_2, \dots, X_p$ : The predictor variables influencing the hazard.

### 5.3.1.Binned Model

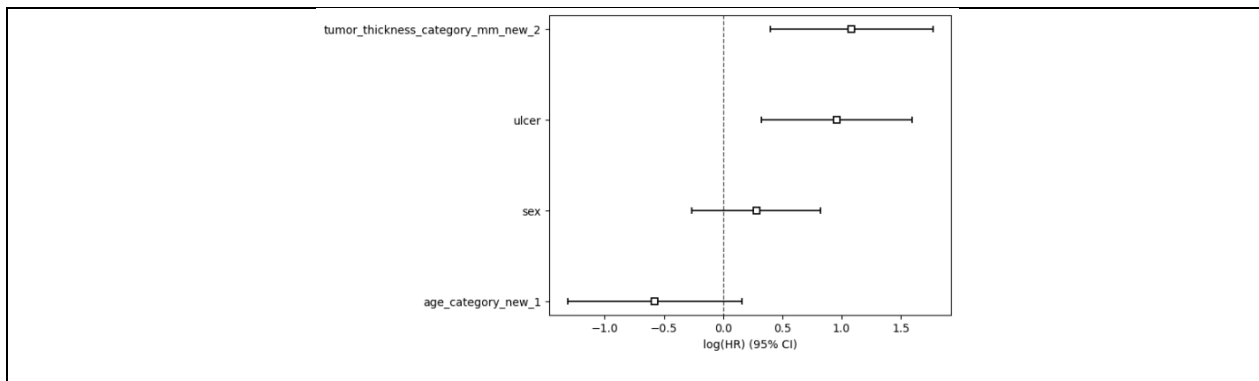
|                                   | coef  | exp(coef) | se(coef) | coef lower 95% | coef upper 95% | exp(coef) lower 95% | exp(coef) upper 95% | cmp to | z     | p      | -log2(p) |
|-----------------------------------|-------|-----------|----------|----------------|----------------|---------------------|---------------------|--------|-------|--------|----------|
| sex                               | 0.28  | 1.32      | 0.28     | -0.27          | 0.82           | 0.77                | 2.28                | 0.00   | 1.00  | 0.32   | 1.67     |
| ulcer                             | 0.96  | 2.61      | 0.32     | 0.33           | 1.60           | 1.38                | 4.94                | 0.00   | 2.96  | <0.005 | 8.36     |
| age_category_new_1                | -0.58 | 0.56      | 0.38     | -1.31          | 0.16           | 0.27                | 1.18                | 0.00   | -1.53 | 0.13   | 2.99     |
| tumor_thickness_category_mm_new_2 | 1.09  | 2.96      | 0.35     | 0.40           | 1.77           | 1.49                | 5.87                | 0.00   | 3.10  | <0.005 | 9.04     |

|                                  |                      |
|----------------------------------|----------------------|
| <b>Concordance</b>               | <b>0.77</b>          |
| <b>Partial AIC</b>               | <b>528.54</b>        |
| <b>log-likelihood ratio test</b> | <b>45.86 on 4 df</b> |
| <b>-log2(p) of ll-ratio test</b> | <b>28.50</b>         |

|                                  |                                |
|----------------------------------|--------------------------------|
| <b>model</b>                     | <b>lifelines.CoxPHFitter</b>   |
| <b>duration col</b>              | <b>'time'</b>                  |
| <b>event col</b>                 | <b>'event'</b>                 |
| <b>baseline estimation</b>       | <b>breslow</b>                 |
| <b>number of observations</b>    | <b>205</b>                     |
| <b>number of events observed</b> | <b>57</b>                      |
| <b>partial log-likelihood</b>    | <b>-260.27</b>                 |
| <b>time fit was run</b>          | <b>2024-12-30 11:11:50 UTC</b> |



**Table 6 – Binned model results.**

**sex:** The HR is 1.32 with a p-value of 0.32. The null hypothesis is not rejected, indicating no significant effect of sex on the hazard.

**ulcer:** The HR is 2.61 with a p-value  $< 0.005$ . The null hypothesis is rejected, indicating that individuals with an ulcer have a significantly higher risk of the event, with a 161% increase in hazard compared to those without an ulcer.

**age\_category\_new\_1:** The HR is 0.56 with a p-value of 0.13. The null hypothesis is not rejected, suggesting no significant effect of this age category on the risk of the event.

**tumor\_thickness\_category\_mm\_new\_2:** The hazard ratio is 2.96 with a p-value  $< 0.005$ . The null hypothesis is rejected, showing that higher tumor thickness significantly increases the risk of the event by 196%.

**AIC:** The model's partial AIC is 528.54, indicating its goodness-of-fit. Lower AIC values are desirable when comparing models.

**Forest plot:** The forest plot shows significant predictors (**ulcer** and **tumor\_thickness\_category\_mm\_new\_2**) with confidence intervals entirely above zero, confirming their strong effects on hazard. Non-significant predictors (**sex** and **age\_category\_new\_1**) have intervals crossing zero, indicating no substantial impact.

#### ➤ Proportional Hazards Assumption Check Binned Model

We choose to make the Proportional Hazard Assumption Check to ensure that the hazard ratio between different groups remains constant over time in models like the Cox Proportional Hazards model. This assumption is critical for the model's validity, as it directly affects the accuracy and reliability of the results and their interpretation. In addition to the proportional hazards assumption, the model relies on several key assumptions: a linear relationship between the covariates and the log hazard, independence of survival times among individuals, and the inclusion of all relevant covariates to avoid bias from omitted variables.

The residuals are randomly scattered around zero, and the p-values for all predictors are greater than 0.05, indicating no significant violation of the assumption (see [Appendix 8.3.1](#)).

### 5.3.2. Continuous Model

|           | coef | exp(coef) | se(coef) | coef lower 95% | coef upper 95% | exp(coef) lower 95% | exp(coef) upper 95% | cmp to | z    | p      | -log2(p) |
|-----------|------|-----------|----------|----------------|----------------|---------------------|---------------------|--------|------|--------|----------|
| sex       | 0.43 | 1.54      | 0.27     | -0.09          | 0.96           | 0.91                | 2.60                | 0.00   | 1.62 | 0.11   | 3.24     |
| age       | 0.01 | 1.01      | 0.01     | -0.00          | 0.03           | 1.00                | 1.03                | 0.00   | 1.47 | 0.14   | 2.82     |
| thickness | 0.11 | 1.12      | 0.04     | 0.03           | 0.18           | 1.04                | 1.20                | 0.00   | 2.89 | <0.005 | 8.01     |
| ulcer     | 1.16 | 3.20      | 0.31     | 0.56           | 1.77           | 1.75                | 5.88                | 0.00   | 3.76 | <0.005 | 12.52    |

|                                  |                      |
|----------------------------------|----------------------|
| <b>Concordance</b>               | <b>0.75</b>          |
| <b>Partial AIC</b>               | <b>532.78</b>        |
| <b>log-likelihood ratio test</b> | <b>41.62 on 4 df</b> |
| <b>-log2(p) of ll-ratio test</b> | <b>25.58</b>         |

|                                  |                         |
|----------------------------------|-------------------------|
| <b>model</b>                     | lifelines.CoxPHFitter   |
| <b>duration col</b>              | 'time'                  |
| <b>event col</b>                 | 'event'                 |
| <b>baseline estimation</b>       | breslow                 |
| <b>number of observations</b>    | 205                     |
| <b>number of events observed</b> | 57                      |
| <b>partial log-likelihood</b>    | -262.39                 |
| <b>time fit was run</b>          | 2024-12-30 11:11:52 UTC |

Table 7 - Continuous model results.

**sex:** The hazard ratio (HR) is 1.54 with a p-value of 0.11. The null hypothesis is not rejected, indicating no significant effect of sex on the hazard.

**age:** The hazard ratio (HR) is 1.01 with a p-value of 0.14. The null hypothesis is not rejected, suggesting no significant effect of age on the hazard.

**thickness:** The hazard ratio (HR) is 1.12 with a p-value < 0.005. The null hypothesis is rejected, indicating that higher thickness significantly increases the risk of the event by 12%.

**ulcer:** The hazard ratio (HR) is 3.20 with a p-value < 0.005. The null hypothesis is rejected, showing that individuals with an ulcer have a significantly higher risk of the event, with a 220% increase in hazard compared to those without an ulcer.

**AIC:** The model's partial AIC is 532.78. Comparing to the Binned model, this model's AIC score is higher, which indicating on poorer results of this model.

**Forest Plot:** The forest plot shows that predictors like **ulcer** and **thickness** are significant because their confidence intervals are above zero, meaning they have a clear effect on the hazard. Predictors like **sex** and **age** are not significant because their confidence intervals cross zero, meaning they don't have a strong or clear impact.

## ➤ Proportional Hazards Assumption Check Continuous Model

The residuals are randomly scattered around zero, and the p-values for all predictors are greater than 0.05, indicating no significant violation of the assumption (see [Appendix 8.3.2](#)).

### 5.4.Question 4

In this stepwise model selection process, we applied both backward elimination and forward elimination to refine the Cox Proportional Hazards (CPH) model after deciding to continue with the binning model because of the lower AIC score gained from the model.

The backward elimination procedure involved starting with a model that included all covariates and systematically removing variables with p-values greater than 0.05. The model was iteratively fitted, and at each step, the variable with the highest p-value was removed if it exceeded the significance threshold. In this case, variables such as **sex** and **age\_category\_new\_1** were removed due to their p-values of 0.315 and 0.070, respectively, which are both above the 0.05 threshold. After this elimination process, the final model retained two variables:

**tumor\_thickness\_category\_mm\_new\_2** and **ulcer**. These selected variables are considered the most significant predictors of survival in this dataset, highlighting the importance of tumour thickness and the presence of ulceration in influencing survival outcomes.

In the forward elimination process, we started with an empty model and sequentially added variables based on their statistical significance. At each step, the variable with the lowest p-value (below the 0.05 threshold) was added to the model, provided it significantly improved the model fit.

Interestingly, the forward elimination approach yielded the same final set of variables as the backward elimination method: **tumor\_thickness\_category\_mm\_new\_2** and **ulcer**.

| Backward elimination                                                                                                                                  | Forward elimination                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Removing sex (p-value = 0.315)<br>Removing age_category_new_1 (p-value = 0.070)<br>Selected variables: ['tumor_thickness_category_mm_new_2', 'ulcer'] | Adding tumor_thickness_category_mm_new_2 (p-value = 0.000)<br>Adding ulcer (p-value = 0.002)<br>Selected variables: ['tumor_thickness_category_mm_new_2', 'ulcer'] |

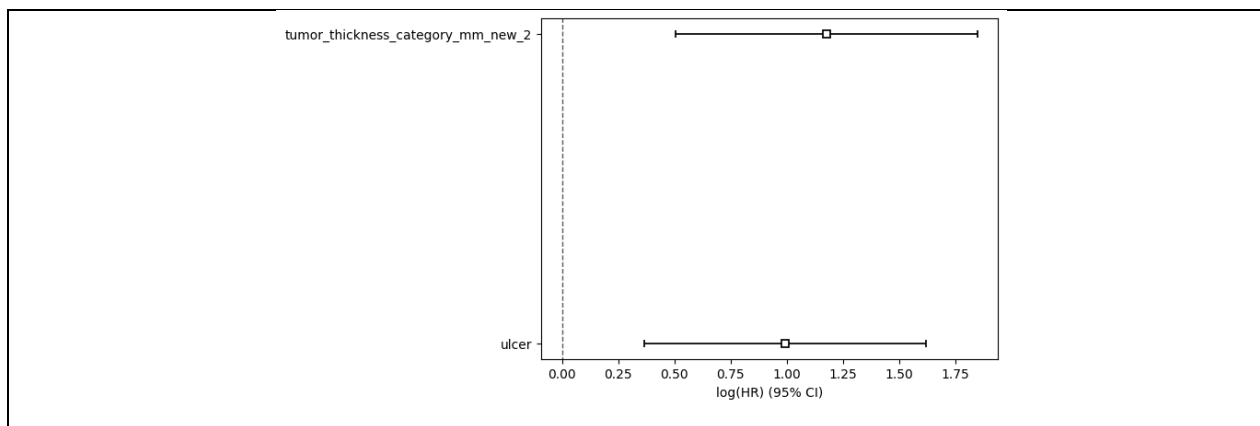
Table 8 - Chosen variables in backward elimination and forward elimination.

#### 5.4.1. Binned Model After Elimination

|                                   | coef | exp(coef) | se(coef) | coef lower 95% | coef upper 95% | exp(coef) lower 95% | exp(coef) upper 95% | cmp to | z    | p      | -log2(p) |
|-----------------------------------|------|-----------|----------|----------------|----------------|---------------------|---------------------|--------|------|--------|----------|
| ulcer                             | 0.99 | 2.69      | 0.32     | 0.36           | 1.62           | 1.44                | 5.04                | 0.00   | 3.10 | <0.005 | 9.00     |
| tumor_thickness_category_mm_new_2 | 1.18 | 3.24      | 0.34     | 0.51           | 1.85           | 1.66                | 6.33                | 0.00   | 3.44 | <0.005 | 10.74    |

|                                                                                                                                                  |                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Concordance</b> 0.74<br><b>Partial AIC</b> 528.37<br><b>log-likelihood ratio test</b> 42.03 on 2 df<br><b>-log2(p) of ll-ratio test</b> 30.32 | <b>model</b> lifelines.CoxPHFitter<br><b>duration col</b> 'time'<br><b>event col</b> 'event'<br><b>baseline estimation</b> breslow<br><b>number of observations</b> 205<br><b>number of events observed</b> 57<br><b>partial log-likelihood</b> -262.19<br><b>time fit was run</b> 2025-01-11 06:41:16 UTC |
|--------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



**Table 9 - Binned model after elimination results.**

**ulcer:** The hazard ratio (HR) is 2.69 with a p-value < 0.005. The null hypothesis is rejected, showing that individuals with an ulcer have a significantly higher risk of the event, with a 169% increase in hazard compared to those without an ulcer.

**tumor\_thickness\_category\_mm\_new\_2:** The hazard ratio (HR) is 3.24 with a p-value < 0.005. The null hypothesis is rejected, indicating that higher tumor thickness significantly increases the risk of the event by 224%.

**AIC:** The model's partial AIC is **528.37**, indicating its goodness-of-fit. Its lower from the original model (before elimination).

**Forest Plot:** The forest plot shows that predictors like **ulcer** and **tumor\_thickness\_category\_mm\_new\_2** are significant because their confidence intervals are above zero, meaning they have a clear effect on the hazard.

The updated binned model is better than the original because it has a slightly lower AIC (**528.37**), showing a better fit. It also focuses on the key predictors, **tumor\_thickness\_category\_mm\_new\_2** and **ulcer**, which are both highly significant with p-values < 0.005. This makes the model more suitable depending on our criteria (AIC).

#### ➤ **Proportional Hazards Assumption Check Binned Model After Elimination**

The residuals are randomly scattered around zero, and the p-values for all predictors are greater than 0.05, indicating no significant violation of the assumption (see [Appendix 8.3.3](#)).

### **5.5. Question 5**

We chose to examine the influence of including interaction variables in the model to observe the complex relationships between predictors.

A model that includes interaction variables may reveal nuanced relationships that might be missed by main effects alone, potentially improving the model's predictive power and explanatory capacity, allows more realistic representation (real-world phenomena often involve complex interplays between variables. Interaction terms help to model these intricate relationships more accurately), get deeper insights by uncovering how different factors work together to influence the outcome, and avoiding misspecification caused by omitting significant interactions.

We ran a Cox Proportional Hazards model to analyze time-to-event data with multiple variables and their interactions, including sex, age, tumor thickness, ulcer status, and various interaction terms. The results showed that the model's performance was not satisfactory, as indicated by a very high AIC

value of **592.96**, reflecting poor model fit. This makes sense because adding interaction terms increases the complexity of the model, which can lead to overfitting, capturing noise rather than real patterns. Additionally, the increased complexity can reduce the statistical power of individual predictors and obscure their effects, further contributing to the poor fit. Simplifying the model by removing unnecessary interaction terms may help reduce the AIC and improve interpretability, leading to a better-fitting model.

### 5.5.1. Full Interaction Model

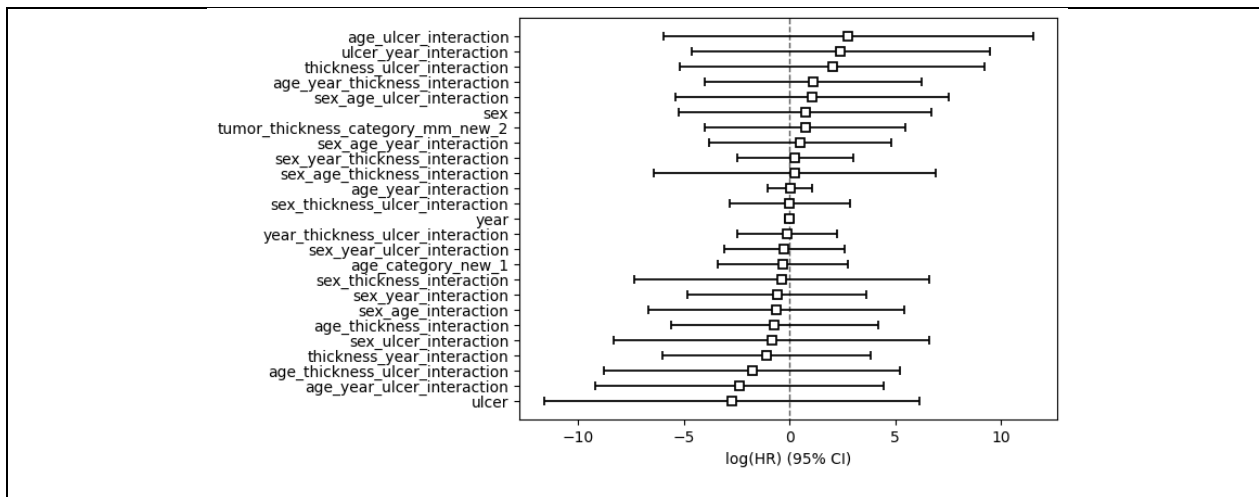
|                                   | coef  | exp(coef) | se(coef) | coef lower 95% | coef upper 95% | exp(coef) lower 95% | exp(coef) upper 95% | cmp to | z     | p    | -log2(p) |
|-----------------------------------|-------|-----------|----------|----------------|----------------|---------------------|---------------------|--------|-------|------|----------|
| sex                               | 0.75  | 2.11      | 3.05     | -5.23          | 6.73           | 0.01                | 837.30              | 0.00   | 0.25  | 0.81 | 0.31     |
| age_category_new_1                | -0.32 | 0.73      | 1.56     | -3.38          | 2.75           | 0.03                | 15.62               | 0.00   | -0.20 | 0.84 | 0.25     |
| year                              | -0.01 | 0.99      | 0.11     | -0.23          | 0.21           | 0.80                | 1.23                | 0.00   | -0.09 | 0.93 | 0.10     |
| tumor_thickness_category_mm_new_2 | 0.74  | 2.09      | 2.42     | -4.00          | 5.47           | 0.02                | 238.02              | 0.00   | 0.31  | 0.76 | 0.40     |
| ulcer                             | -2.74 | 0.06      | 4.53     | -11.63         | 6.14           | 0.00                | 463.20              | 0.00   | -0.61 | 0.54 | 0.88     |
| sex_age_interaction               | -0.64 | 0.53      | 3.09     | -6.70          | 5.41           | 0.00                | 223.46              | 0.00   | -0.21 | 0.83 | 0.26     |
| sex_year_interaction              | -0.60 | 0.55      | 2.15     | -4.81          | 3.62           | 0.01                | 37.26               | 0.00   | -0.28 | 0.78 | 0.36     |
| sex_thickness_interaction         | -0.37 | 0.69      | 3.56     | -7.35          | 6.60           | 0.00                | 736.84              | 0.00   | -0.10 | 0.92 | 0.13     |
| sex_ulcer_interaction             | -0.85 | 0.43      | 3.82     | -8.33          | 6.63           | 0.00                | 757.87              | 0.00   | -0.22 | 0.82 | 0.28     |
| age_year_interaction              | 0.02  | 1.02      | 0.54     | -1.04          | 1.08           | 0.35                | 2.93                | 0.00   | 0.03  | 0.98 | 0.03     |
| age_thickness_interaction         | -0.70 | 0.49      | 2.50     | -5.60          | 4.19           | 0.00                | 65.98               | 0.00   | -0.28 | 0.78 | 0.36     |
| age_ulcer_interaction             | 2.77  | 15.91     | 4.46     | -5.98          | 11.51          | 0.00                | 1.00e+05            | 0.00   | 0.62  | 0.54 | 0.90     |
| thickness_year_interaction        | -1.09 | 0.34      | 2.53     | -6.04          | 3.86           | 0.00                | 47.54               | 0.00   | -0.43 | 0.67 | 0.58     |
| thickness_ulcer_interaction       | 2.03  | 7.62      | 3.67     | -5.17          | 9.23           | 0.01                | 10227.51            | 0.00   | 0.55  | 0.58 | 0.78     |
| ulcer_year_interaction            | 2.42  | 11.28     | 3.61     | -4.65          | 9.50           | 0.01                | 13308.08            | 0.00   | 0.67  | 0.50 | 0.99     |
| sex_age_year_interaction          | 0.51  | 1.66      | 2.19     | -3.78          | 4.80           | 0.02                | 121.84              | 0.00   | 0.23  | 0.82 | 0.29     |
| sex_age_thickness_interaction     | 0.24  | 1.27      | 3.41     | -6.44          | 6.92           | 0.00                | 1008.49             | 0.00   | 0.07  | 0.94 | 0.08     |
| sex_age_ulcer_interaction         | 1.06  | 2.87      | 3.29     | -5.40          | 7.51           | 0.00                | 1825.13             | 0.00   | 0.32  | 0.75 | 0.42     |
| sex_year_thickness_interaction    | 0.27  | 1.30      | 1.41     | -2.49          | 3.03           | 0.08                | 29.60               | 0.00   | 0.19  | 0.85 | 0.23     |
| sex_year_ulcer_interaction        | -0.26 | 0.77      | 1.45     | -3.10          | 2.58           | 0.05                | 13.26               | 0.00   | -0.18 | 0.86 | 0.22     |
| sex_thickness_ulcer_interaction   | -0.00 | 1.00      | 1.45     | -2.85          | 2.85           | 0.06                | 17.22               | 0.00   | -0.00 | 1.00 | 0.00     |
| age_year_thickness_interaction    | 1.11  | 3.05      | 2.61     | -4.01          | 6.24           | 0.02                | 512.48              | 0.00   | 0.43  | 0.67 | 0.58     |
| age_year_ulcer_interaction        | -2.37 | 0.09      | 3.48     | -9.20          | 4.46           | 0.00                | 86.12               | 0.00   | -0.68 | 0.50 | 1.01     |
| age_thickness_ulcer_interaction   | -1.78 | 0.17      | 3.58     | -8.79          | 5.23           | 0.00                | 186.89              | 0.00   | -0.50 | 0.62 | 0.69     |
| year_thickness_ulcer_interaction  | -0.11 | 0.89      | 1.20     | -2.46          | 2.24           | 0.09                | 9.38                | 0.00   | -0.09 | 0.93 | 0.11     |

|                           |                |
|---------------------------|----------------|
| Concordance               | 0.81           |
| Partial AIC               | 592.96         |
| log-likelihood ratio test | 23.44 on 25 df |
| -log2(p) of ll-ratio test | 0.86           |

|                           |                         |
|---------------------------|-------------------------|
| model                     | lifelines.CoxPHFitter   |
| duration col              | 'time'                  |
| event col                 | 'event'                 |
| baseline estimation       | breslow                 |
| number of observations    | 205                     |
| number of events observed | 57                      |
| partial log-likelihood    | -271.48                 |
| time fit was run          | 2025-01-11 06:41:18 UTC |



**Table 10 – Full interaction model.**

**age\_category\_new\_1:** The hazard ratio (HR) is 0.73 with a p-value of 0.84. The null hypothesis is not rejected, showing no significant effect of age category on the hazard.

**year:** The hazard ratio (HR) is 0.99 with a p-value of 0.93. The null hypothesis is not rejected, meaning that the year does not significantly influence the hazard.

**tumor\_thickness\_category\_mm\_new\_2:** The hazard ratio (HR) is 2.09 with a p-value of 0.76. The null hypothesis is not rejected, but the HR suggests a potential 172% increase in risk, though this effect is not statistically significant.

**ulcer:** The hazard ratio (HR) is 0.06 with a p-value of 0.54. The null hypothesis is not rejected, indicating no significant effect of ulcer status on the hazard.

**sex\_age\_interaction:** The hazard ratio (HR) is 0.53 with a p-value of 0.83. The null hypothesis is not rejected, suggesting no significant interaction between sex and age.

**sex\_year\_interaction:** The hazard ratio (HR) is 0.55 with a p-value of 0.78. The null hypothesis is not rejected, suggesting no significant interaction between sex and age.

**sex\_thickness\_interaction:** The hazard ratio (HR) is 0.69 with a p-value of 0.92. The null hypothesis is not rejected, showing no significant interaction between sex and tumor thickness.

**sex\_ulcer\_interaction:** The hazard ratio (HR) is 0.43 with a p-value of 0.82. The null hypothesis is not rejected, meaning no significant interaction between sex and ulcer.

**age\_year\_interaction:** The hazard ratio (HR) is 1.02 with a p-value of 0.98. The null hypothesis is not rejected, meaning no significant interaction between sex and ulcer.

**age\_thickness\_interaction:** The hazard ratio (HR) is 0.49 with a p-value of 0.78. The null hypothesis is not rejected, indicating no significant interaction between age and tumor thickness.

**age\_ulcer\_interaction:** The hazard ratio (HR) is 15.91 with a p-value of 0.54. The null hypothesis is not rejected, suggesting no significant interaction between age and ulcer.

**thickness\_year\_interaction:** The hazard ratio (HR) is 0.34 with a p-value of 0.67. The null hypothesis is not rejected, indicating no significant interaction between tumor thickness and year.

**thickness\_ulcer\_interaction:** The hazard ratio (HR) is 7.62 with a p-value of 0.58. The null hypothesis is not rejected, indicating no significant interaction between tumor thickness and year.

**ulcer\_year\_interaction:** The hazard ratio (HR) is 11.28 with a p-value of 0.5. The null hypothesis is not rejected, meaning no significant interaction between ulcer and year.

**sex\_age\_year\_interaction:** The hazard ratio (HR) is 1.66 with a p-value of 0.82. The null hypothesis is not rejected, indicating no significant three-way interaction.

**sex\_age\_thickness\_interaction:** The hazard ratio (HR) is 1.27 with a p-value of 0.94. The null hypothesis is not rejected, indicating no significant three-way interaction.

**sex\_age\_ulcer\_interaction:** The hazard ratio (HR) is 2.87 with a p-value of 0.75. The null hypothesis is not rejected, suggesting no significant three-way interaction between sex, age, and ulcer.

**sex\_year\_thickness\_interaction:** The hazard ratio (HR) is 1.3 with a p-value of 0.85. The null hypothesis is not rejected, indicating no significant three-way interaction between sex, year, and ulcer.

**sex\_year\_ulcer\_interaction:** The hazard ratio (HR) is 0.77 with a p-value of 0.86. The null hypothesis is not rejected, indicating no significant three-way interaction between sex, year, and ulcer.

**sex\_thickness\_ulcer\_interaction:** The hazard ratio (HR) is 1 with a p-value of 1. The null hypothesis is not rejected, indicating no significant three-way interaction between sex, year, and ulcer.

**age\_year\_thickness\_interaction:** The hazard ratio (HR) is 3.05 with a p-value of 0.67. The null hypothesis is not rejected, suggesting no significant three-way interaction between age, year, and tumor thickness.

**age\_thickness\_ulcer\_interaction:** The hazard ratio (HR) is 0.17 with a p-value of 0.62. The null hypothesis is not rejected, showing no significant three-way interaction between age, year, and ulcer.

**age\_year\_ulcer\_interaction:** The hazard ratio (HR) is 0.09 with a p-value of 0.5. The null hypothesis is not rejected, showing no significant three-way interaction between age, year, and ulcer.

**year\_thickness\_ulcer\_interaction:** The hazard ratio (HR) is 0.89 with a p-value of 0.93. The null hypothesis is not rejected, meaning no significant interaction between year, tumor thickness, and ulcer.

**AIC:** The model's AIC is **592.96**, which is very high (comparing to the AIC of the binned model after elimination), indicating a poor model fit. Lower AIC values are desirable when comparing models.

**Forest Plot:** The forest plot shows that none of the predictors are statistically significant because their confidence intervals cross the value of zero.

#### ➤ **Proportional Hazards Assumption Check Full Interactions Model**

The residuals are randomly scattered around zero, and the p-values for all predictors are greater than 0.05, indicating no significant violation of the assumption (see [Appendix 8.3.4](#)).

```

Removing sex_thickness_ulcer_interaction (p-value = 1.000)
Removing sex_age_thickness_interaction (p-value = 0.956)
Removing year_category (p-value = 0.940)
Removing age_year_interaction (p-value = 0.969)
Removing year (p-value = 0.920)
Removing year_thickness_ulcer_interaction (p-value = 0.939)
Removing sex_thickness_interaction (p-value = 0.897)
Removing sex_year_ulcer_interaction (p-value = 0.879)
Removing sex_year_thickness_interaction (p-value = 0.916)
Removing sex_age_interaction (p-value = 0.829)
Removing sex_age_year_interaction (p-value = 0.897)
Removing sex_year_interaction (p-value = 0.817)
Removing sex (p-value = 0.903)
Removing age_thickness_interaction (p-value = 0.738)
Removing tumor_thickness_category_mm_new_2 (p-value = 0.847)
Removing thickness_year_interaction (p-value = 0.577)
Removing age_thickness_ulcer_interaction (p-value = 0.998)
Removing ulcer (p-value = 0.995)
Removing age_year_thickness_interaction (p-value = 0.182)
Removing age_ulcer_interaction (p-value = 0.266)
Removing ulcer_year_interaction (p-value = 0.198)
Removing age_year_ulcer_interaction (p-value = 0.555)
Removing sex_ulcer_interaction (p-value = 0.050)
Removing sex_age_ulcer_interaction (p-value = 0.216)
Removing age_category_new_1 (p-value = 0.065)
Selected variables: ['thickness_ulcer_interaction']

```

**Table 11 - Chosen variables in backward elimination on the full interaction model.**

The backward elimination procedure involved starting with the full interaction model that included all covariates and all the interaction variables from 2<sup>nd</sup> and 3<sup>rd</sup> order. The algorithm systematically removes variables with p-values greater than 0.05. In this case, variables such as **sex\_thickness\_ulcer\_interaction** and **year** were removed due to their p-values of 1.000 and 0.920, respectively, which are both above the 0.05 threshold. After this elimination process, the final model retained one variable: **thickness\_ulcer\_interaction**. This selected variable considered the most significant predictors of survival in this dataset, highlighting the importance of the interaction between the tumour thickness and the presence of ulceration in influencing survival outcomes.

### 5.5.2. Full Interaction Model After Elimination

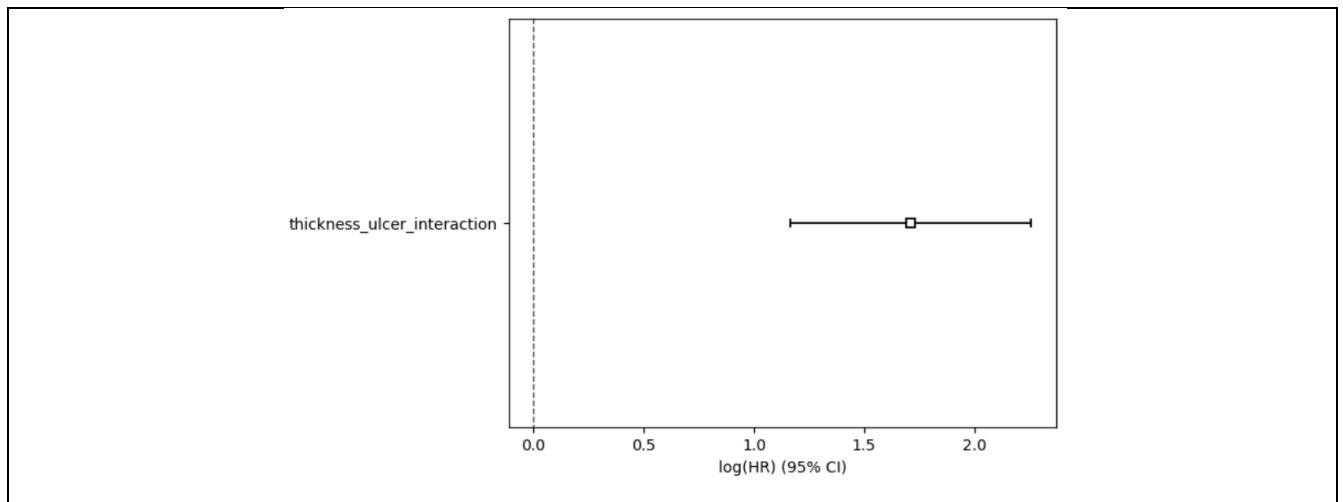
|                             | coef | exp(coef) | se(coef) | coef lower 95% | coef upper 95% | exp(coef) lower 95% | exp(coef) upper 95% | cmp to | z    | p      | -log2(p) |
|-----------------------------|------|-----------|----------|----------------|----------------|---------------------|---------------------|--------|------|--------|----------|
| thickness_ulcer_interaction | 1.71 | 5.54      | 0.28     | 1.17           | 2.26           | 3.21                | 9.57                | 0.00   | 6.14 | <0.005 | 30.19    |

|                                  |               |
|----------------------------------|---------------|
| <b>Concordance</b>               | 0.72          |
| <b>Partial AIC</b>               | 528.61        |
| <b>log-likelihood ratio test</b> | 39.79 on 1 df |
| <b>-log2(p) of ll-ratio test</b> | 31.72         |

|                                  |                       |
|----------------------------------|-----------------------|
| <b>model</b>                     | lifelines.CoxPHFitter |
| <b>duration col</b>              | 'time'                |
| <b>event col</b>                 | 'event'               |
| <b>baseline estimation</b>       | breslow               |
| <b>number of observations</b>    | 205                   |
| <b>number of events observed</b> | 57                    |
| <b>partial log-likelihood</b>    | -263.30               |



**Table 12 - Full interaction model after elimination results.**

**thickness\_ulcer\_interaction:** The hazard ratio (HR) is 5.54, with a p-value that is smaller than 0.005. The null hypothesis is rejected, confirming that this effect is statistically significant.

**AIC:** The partial AIC is **528.61**, indicating a reasonably good fit for the model, compared to the full model before the elimination, and almost similar to the AIC of the binned model after elimination.

**Forest Plot:** The forest plot shows the confidence interval for the **thickness\_ulcer\_interaction** is entirely above zero, confirming a clear and significant positive effect on the hazard.

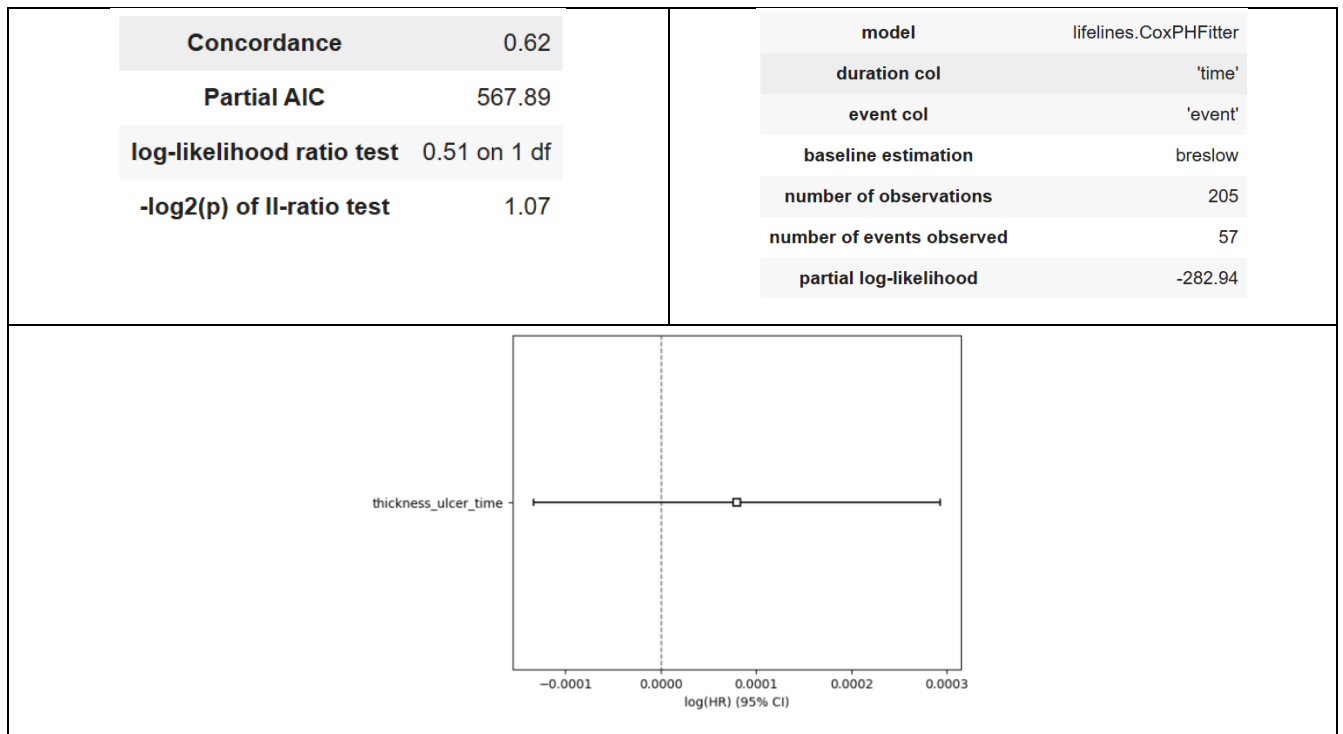
#### ➤ Proportional Hazards Assumption Check Full Interaction Model After Elimination

The p-values for **thickness\_ulcer\_interaction** predictor is less than 0.05, suggesting a significant violation of the proportional hazards assumption for this interaction term (see [Appendix 8.3.5](#)).

We chose to use an approach that addresses the violation of the proportional hazards assumption by allowing the effect of the problematic variable to change over time (Kleinbaum & Klein, 2012). This method involves multiplying the non-proportional variable by the time variable, allowing its effect to vary over time. We chose this approach because it directly tackles the non-proportionality issue identified in the 'thickness\_ulcer\_interaction' term, while preserving the overall structure of the Cox model. By incorporating time-dependency, we aim to improve the model's accuracy and reliability. It allows for a more nuanced and potentially more accurate representation of the variable's impact on survival outcomes across different time points, leading to improved model fit.

#### 5.5.3. Full Interaction Model After Elimination And Adjusting The Variable Over Time

|                      | coef | exp(coef) | se(coef) | coef lower<br>95% | coef upper<br>95% | exp(coef) lower<br>95% | exp(coef) upper<br>95% | cmp<br>to | z    | p    | -<br>log2(p) |
|----------------------|------|-----------|----------|-------------------|-------------------|------------------------|------------------------|-----------|------|------|--------------|
| thickness_ulcer_time | 0.00 | 1.00      | 0.00     | -0.00             | 0.00              | 1.00                   | 1.00                   | 0.00      | 0.73 | 0.46 | 1.11         |



**Table 13 - Full interaction model after elimination results and adjusting the variable over time.**

**thickness\_ulcer\_time:** The hazard ratio (HR) is 1.00 with a p-value of 0.46. The null hypothesis is not rejected, indicating that the interaction between thickness and ulcer does not significantly affect the risk of the event. The hazard ratio of 1.00 suggests that this interaction term, when adjusted for time, has essentially no effect on the hazard rate.

**AIC:** The partial AIC is **567.89**, which is very high (comparing to the AIC of the binned model after elimination), indicating a poor model fit. Lower AIC values are desirable when comparing models.

**Forest Plot:** The forest plot shows that the predictor is not statistically significant because its confidence intervals cross the value of zero.

### ➤ Proportional Hazards Assumption Check Full Interaction Model After Elimination And Adjusting The Variable Over Time

The residuals are randomly scattered around zero, and the p-values for all predictors are greater than 0.05, indicating no significant violation of the assumption (see [Appendix 8.3.6](#)). It is important to note that although there is no significant violation of the assumption, the interaction between thickness and ulcer does not appear to have a meaningful impact on the hazard rate over time.

#### 5.5.4. Partial Interaction Model

|                                   | coef | exp(coef) | se(coef) | coef  | lower 95% | coef | upper 95% | exp(coef) | lower 95% | exp(coef) | upper 95% | cmp to | z    | p    | log2(p) |
|-----------------------------------|------|-----------|----------|-------|-----------|------|-----------|-----------|-----------|-----------|-----------|--------|------|------|---------|
| tumor_thickness_category_mm_new_2 | 0.86 | 2.37      | 0.50     | -0.13 |           | 1.85 |           | 0.88      |           | 6.37      |           | 0.00   | 1.71 | 0.09 | 3.52    |
| ulcer                             | 0.57 | 1.77      | 0.60     | -0.61 |           | 1.75 |           | 0.55      |           | 5.75      |           | 0.00   | 0.95 | 0.34 | 1.53    |
| thickness_ulcer_interaction       | 0.62 | 1.86      | 0.73     | -0.81 |           | 2.05 |           | 0.44      |           | 7.76      |           | 0.00   | 0.85 | 0.40 | 1.33    |

|                                                                                                                                                  |                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Concordance</b> 0.74<br><b>Partial AIC</b> 529.62<br><b>log-likelihood ratio test</b> 42.78 on 3 df<br><b>-log2(p) of ll-ratio test</b> 28.44 | <b>model</b> lifelines.CoxPHFitter<br><b>duration col</b> 'time'<br><b>event col</b> 'event'<br><b>baseline estimation</b> breslow<br><b>number of observations</b> 205<br><b>number of events observed</b> 57<br><b>partial log-likelihood</b> -261.81<br><b>time fit was run</b> 2025-01-11 06:41:19 UTC |
|--------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

The forest plot displays the log(Hazard Ratio) and 95% confidence intervals for three predictors. The x-axis is labeled 'log(HR) (95% CI)' and ranges from -0.5 to 2.0. A vertical dashed line at 0.0 represents the null hypothesis. The predictors and their approximate values are:

| Predictor                         | log(HR) | 95% CI Lower | 95% CI Upper |
|-----------------------------------|---------|--------------|--------------|
| tumor_thickness_category_mm_new_2 | 0.8     | -0.1         | 1.7          |
| thickness_ulcer_interaction       | 0.6     | -0.2         | 1.4          |
| ulcer                             | 0.6     | -0.1         | 1.3          |

**Table 14 – partial interaction model.**

This model is based on the model that was obtained in [question 4](#) (binned model after elimination).

**Tumor\_thickness\_category\_mm\_new\_2:** The hazard ratio (HR) is 2.37 with a p-value of 0.09. The null hypothesis is not rejected, meaning there is no significant effect. However, the relatively low p-value suggests a trend towards increased risk.

**ulcer:** The hazard ratio (HR) is 1.77 with a p-value of 0.34. The null hypothesis is not rejected, indicating no significant effect.

**thickness\_ulcer\_interaction:** The hazard ratio (HR) is 1.86 with a p-value of 0.40. The null hypothesis is not rejected, meaning no significant interaction between tumor thickness and ulcer presence.

**AIC:** The model's AIC is **592.96**, which is very high, indicating a poor model fit. Lower AIC values are desirable when comparing models.

**Forest Plot:** The forest plot shows that none of the predictors are statistically significant because their confidence intervals cross the value of zero.

#### ➤ **Proportional Hazards Assumption Partial Interactions Model**

The residuals are randomly scattered around zero, and the p-values for all predictors are greater than 0.05, indicating no significant violation of the assumption (see [Appendix 8.3.7](#)).

Removing thickness\_ulcer\_interaction (p-value = 0.397)  
 Selected variables: ['tumor\_thickness\_category\_mm\_new\_2', 'ulcer']

**Table 15 - Chosen variables in backward elimination on the partial interaction model.**

The backward elimination procedure involved starting with the partial interaction model that included the tumour thickness and the presence of ulceration covariates and the interaction between those two variables. The algorithm systematically removes variables with p-values greater than 0.05. In this case, the variable that was removed is **thickness\_ulcer\_interaction** due to its p-values of 0.397, which is above the 0.05 threshold. After this elimination process, the final model retained two variables: **tumor\_thickness\_category\_mm\_new\_2** and **ulcer**. This selected variable considered the most significant predictors of survival in this dataset, highlighting the importance of the tumour thickness and the presence of ulceration in influencing survival outcomes.

### 5.5.5. Partial Interaction Model After Elimination

|                                   | coef | exp(coef) | se(coef) | coef lower 95% | coef upper 95% | exp(coef) lower 95% | exp(coef) upper 95% | cmp to | z    | p      | -log2(p) |
|-----------------------------------|------|-----------|----------|----------------|----------------|---------------------|---------------------|--------|------|--------|----------|
| ulcer                             | 0.99 | 2.69      | 0.32     | 0.36           | 1.62           | 1.44                | 5.04                | 0.00   | 3.10 | <0.005 | 9.00     |
| tumor_thickness_category_mm_new_2 | 1.18 | 3.24      | 0.34     | 0.51           | 1.85           | 1.66                | 6.33                | 0.00   | 3.44 | <0.005 | 10.74    |

|                                  |               |
|----------------------------------|---------------|
| <b>Concordance</b>               | 0.74          |
| <b>Partial AIC</b>               | 528.37        |
| <b>log-likelihood ratio test</b> | 42.03 on 2 df |
| <b>-log2(p) of ll-ratio test</b> | 30.32         |

|                                  |                       |
|----------------------------------|-----------------------|
| <b>model</b>                     | lifelines.CoxPHFitter |
| <b>duration col</b>              | 'time'                |
| <b>event col</b>                 | 'event'               |
| <b>baseline estimation</b>       | breslow               |
| <b>number of observations</b>    | 205                   |
| <b>number of events observed</b> | 57                    |
| <b>partial log-likelihood</b>    | -262.19               |

**Table 16 - Partial interaction model after elimination results.**

After backward elimination we got the same model as the binned model after elimination that obtained from question 4, and its parameters, as well as proportional Hazards assumption check, were already analysed ([link](#)).

#### ➤ Proportional Hazards Assumption Check Partial Interaction Model After Elimination

The residuals are randomly scattered around zero, and the p-values for all predictors are greater than 0.05, indicating no significant violation of the assumption (see [Appendix 8.3.8](#)).

### **5.6. Question 6**

A log-likelihood chi-square test was not conducted to compare the full interaction model and the partial interaction model, as the full interaction model contains only one variable that lacks statistical significance. Given this, the additional complexity of the full interaction model does not provide meaningful improvement in explaining the hazard rate. The partial interaction model, focusing on statistically significant variables, is sufficient for understanding the data, making the test unnecessary.

## 6. Discussion & Conclusions

The analysis of malignant melanoma survival data highlighted the significant impact of tumor thickness and the presence of ulceration on patient outcomes. These factors emerged as the most critical predictors, with thicker tumors and ulcerated lesions significantly decreasing survival probabilities. While Kaplan-Meier curves reveals trends of higher survival probabilities among females and middle-aged adults, potentially reflecting biological and behavioral differences suggested differences in survival based on age and sex, these variables were not statistically significant in the refined Cox models. Although that interaction variables can demonstrated a complex interplay of the data, the analysis of interaction terms in this data was not statistically significant. This may be resulted in the fact that there is no complex connection between the variables. Another theory that can explain this is the fact that the data is censored, or the small amount of observations.

The chosen model is the "Binned Model After Elimination" based on its AIC score (528.37). This model was chosen over the continuous model and the interaction models. This may indicate that the tiny differences between the continuous variable values do not add value to the model, and the binning methodology of tumor thickness captures the relevant information from the variable. Nevertheless, the interaction between the variables of the models does not add relevant information to the predicting of the chances of survival. From this model, it can be shown that the presence of an ulcer and its thickness has the most contribution to the prediction of the chances of survival of a melanoma patient.

Given the reliance on historical data from 1962–1977, the findings must be interpreted within their temporal context. Advances in diagnostic methods, treatment modalities, and healthcare access since then are likely to influence modern melanoma survival outcomes. Despite these limitations, the study underscores the importance of early detection and targeted interventions, as the strong association between thinner tumors and higher survival probabilities highlights how identifying and treating melanoma at earlier stages can substantially improve survival probability. Moving forward, research should leverage larger dataset and contemporary datasets to explore additional predictors, such as genomic factors, lifestyle factors, and socioeconomic factors.

## 7. Bibliography

- Baade, P. D., Whiteman, D. C., Janda, M., Cust, A. E., Neale, R. E., Smithers, B. M., Green, A. C., Khosrotehrani, K., Mar, V., Soyer, H. P., & Aitken, J. F. (2020). Long-term deaths from melanoma according to tumor thickness at diagnosis. *International Journal of Cancer*, 147(5), 1391–1396. <https://doi.org/10.1002/IJC.32930>
- Kleinbaum, D. G., & Klein, M. (2012). *Extension of the Cox Proportional Hazards Model for Time-Dependent Variables*. 241–288. [https://doi.org/10.1007/978-1-4419-6646-9\\_6](https://doi.org/10.1007/978-1-4419-6646-9_6)
- Stidham, K. R., Johnson, J. L., & Seigler, H. F. (n.d.). *Survival Superiority of Females With Melanoma A Multivariate Analysis of 6383 Patients Exploring the Significance of Gender in Prognostic Outcome*.

## 8. Appendixes

### 8.1. Appendix 1 – Pairwise log rank age category inner groups differences

```
Pairwise Log-Rank Tests for age_category:
75+ vs 55-74: p-value = 0.0479
Significant difference between 75+ and 55-74.
75+ vs 35-54: p-value = 0.0021
Significant difference between 75+ and 35-54.
75+ vs 18-34: p-value = 0.0044
Significant difference between 75+ and 18-34.
75+ vs 0-17: p-value = 0.4635
No significant difference between 75+ and 0-17.
55-74 vs 35-54: p-value = 0.2155
No significant difference between 55-74 and 35-54.
55-74 vs 18-34: p-value = 0.1484
No significant difference between 55-74 and 18-34.
55-74 vs 0-17: p-value = 0.6780
No significant difference between 55-74 and 0-17.
35-54 vs 18-34: p-value = 0.5100
No significant difference between 35-54 and 18-34.
35-54 vs 0-17: p-value = 0.3871
No significant difference between 35-54 and 0-17.
18-34 vs 0-17: p-value = 0.2437
No significant difference between 18-34 and 0-17.
```

(Back to [age category variable Pairwise Log-Rank test results](#))

### 8.2. Appendix 2 – pairwise log rank thickness category inner groups differences

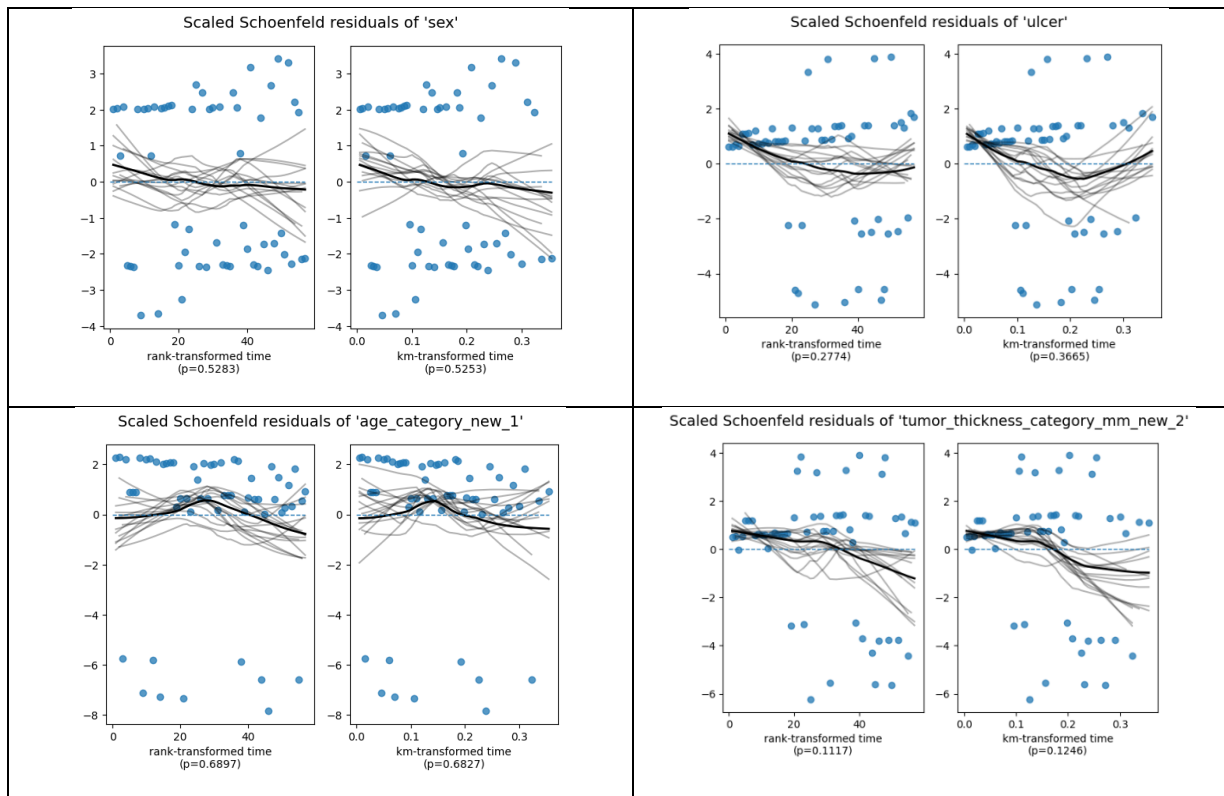
```
Pairwise Log-Rank Tests for tumor_thickness_category_mm:
>6.0 vs 0-0.8: p-value = 0.0003
Significant difference between >6.0 and 0-0.8.
>6.0 vs 1.0-2.0: p-value = 0.0039
Significant difference between >6.0 and 1.0-2.0.
>6.0 vs 2.0-3.0: p-value = 0.7538
No significant difference between >6.0 and 2.0-3.0.
>6.0 vs 4.0-5.0: p-value = 0.2795
No significant difference between >6.0 and 4.0-5.0.
>6.0 vs 5.0-6.0: p-value = 0.3876
No significant difference between >6.0 and 5.0-6.0.
>6.0 vs 3.0-4.0: p-value = 0.9180
No significant difference between >6.0 and 3.0-4.0.
>6.0 vs 0.8-1.0: p-value = 0.1289
No significant difference between >6.0 and 0.8-1.0.
0-0.8 vs 1.0-2.0: p-value = 0.0965
No significant difference between 0-0.8 and 1.0-2.0.
0-0.8 vs 2.0-3.0: p-value = 0.0003
Significant difference between 0-0.8 and 2.0-3.0.
0-0.8 vs 4.0-5.0: p-value = 0.0000
Significant difference between 0-0.8 and 4.0-5.0.
0-0.8 vs 5.0-6.0: p-value = 0.0000
Significant difference between 0-0.8 and 5.0-6.0.
0-0.8 vs 3.0-4.0: p-value = 0.0001
Significant difference between 0-0.8 and 3.0-4.0.
0-0.8 vs 0.8-1.0: p-value = 0.0207
Significant difference between 0-0.8 and 0.8-1.0.
1.0-2.0 vs 2.0-3.0: p-value = 0.0070
Significant difference between 1.0-2.0 and 2.0-3.0.
1.0-2.0 vs 4.0-5.0: p-value = 0.0000
Significant difference between 1.0-2.0 and 4.0-5.0.
1.0-2.0 vs 5.0-6.0: p-value = 0.0001
Significant difference between 1.0-2.0 and 5.0-6.0.
1.0-2.0 vs 3.0-4.0: p-value = 0.0023
Significant difference between 1.0-2.0 and 3.0-4.0.
1.0-2.0 vs 0.8-1.0: p-value = 0.3409
No significant difference between 1.0-2.0 and 0.8-1.0.
2.0-3.0 vs 4.0-5.0: p-value = 0.0919
No significant difference between 2.0-3.0 and 4.0-5.0.
2.0-3.0 vs 5.0-6.0: p-value = 0.1659
No significant difference between 2.0-3.0 and 5.0-6.0.
2.0-3.0 vs 3.0-4.0: p-value = 0.8424
No significant difference between 2.0-3.0 and 3.0-4.0.
2.0-3.0 vs 0.8-1.0: p-value = 0.1949
No significant difference between 2.0-3.0 and 0.8-1.0.
4.0-5.0 vs 5.0-6.0: p-value = 0.9024
No significant difference between 4.0-5.0 and 5.0-6.0.
4.0-5.0 vs 3.0-4.0: p-value = 0.0773
No significant difference between 4.0-5.0 and 3.0-4.0.
4.0-5.0 vs 0.8-1.0: p-value = 0.0048
Significant difference between 4.0-5.0 and 0.8-1.0.
5.0-6.0 vs 3.0-4.0: p-value = 0.1342
No significant difference between 5.0-6.0 and 3.0-4.0.
5.0-6.0 vs 0.8-1.0: p-value = 0.0087
Significant difference between 5.0-6.0 and 0.8-1.0.
3.0-4.0 vs 0.8-1.0: p-value = 0.1849
No significant difference between 3.0-4.0 and 0.8-1.0.
```

(Back to [Thickness category variable Pairwise Log-Rank test results](#))

### 8.3. Appendix 3 – Proportional Hazards Assumption Check

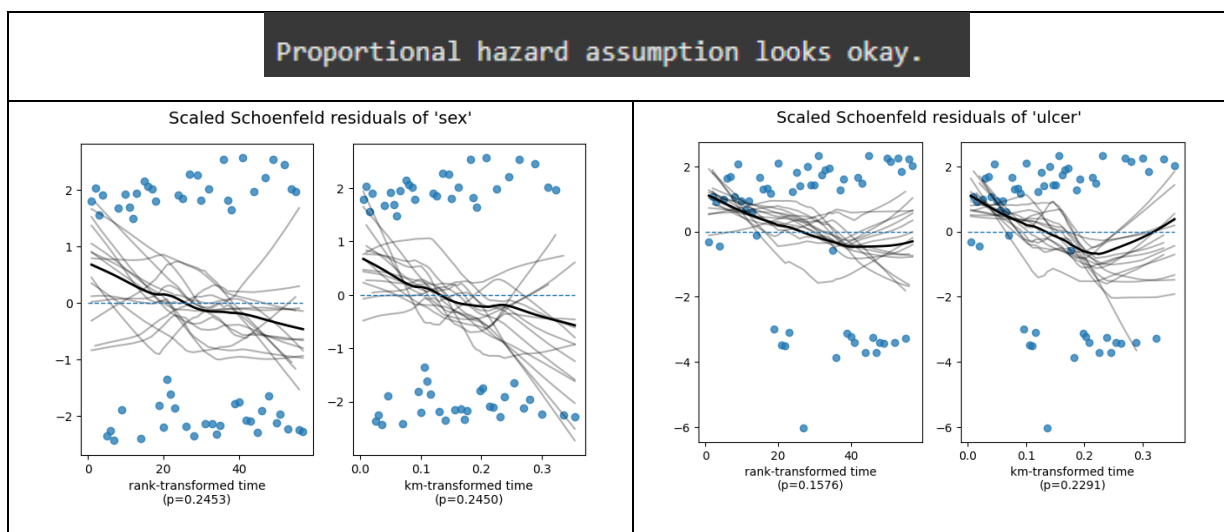
#### 8.3.1. Assumption check of the binned model

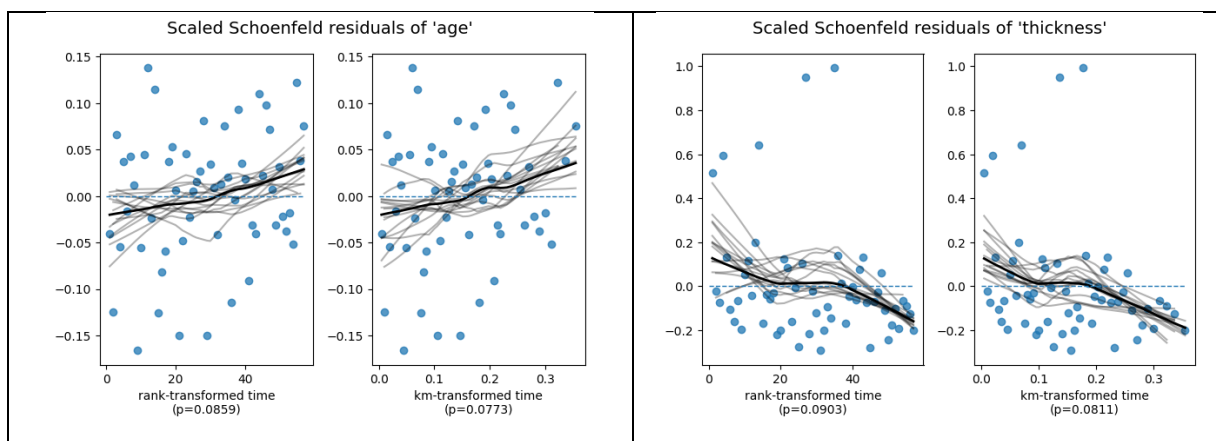
Proportional hazard assumption looks okay.



(Back to [Proportional Hazards Assumption Check Binned Model](#))

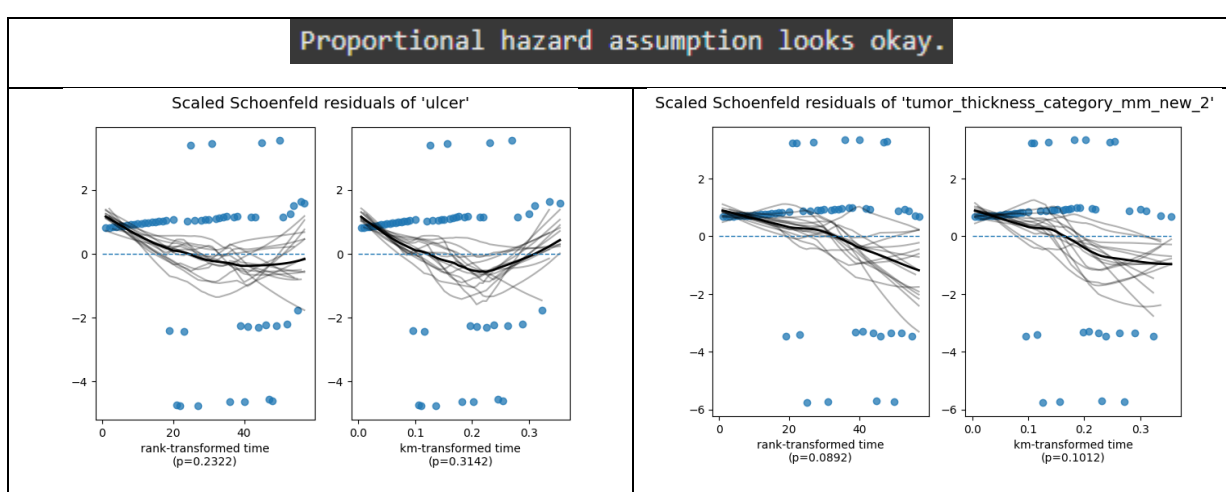
### 8.3.2.Assumption check continuous model





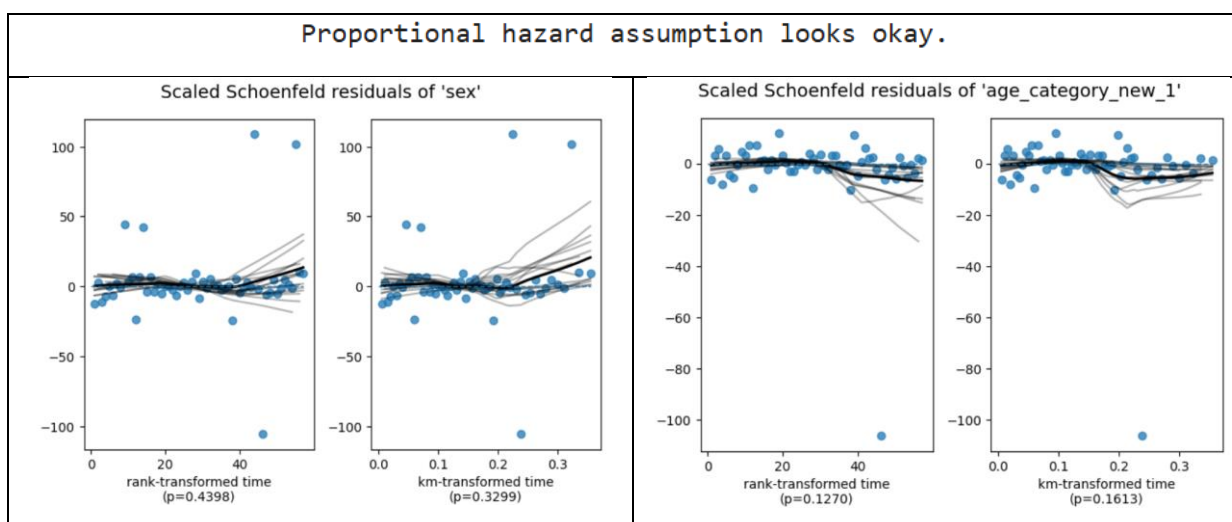
(Back to [Proportional Hazards Assumption Check Continuous Model](#))

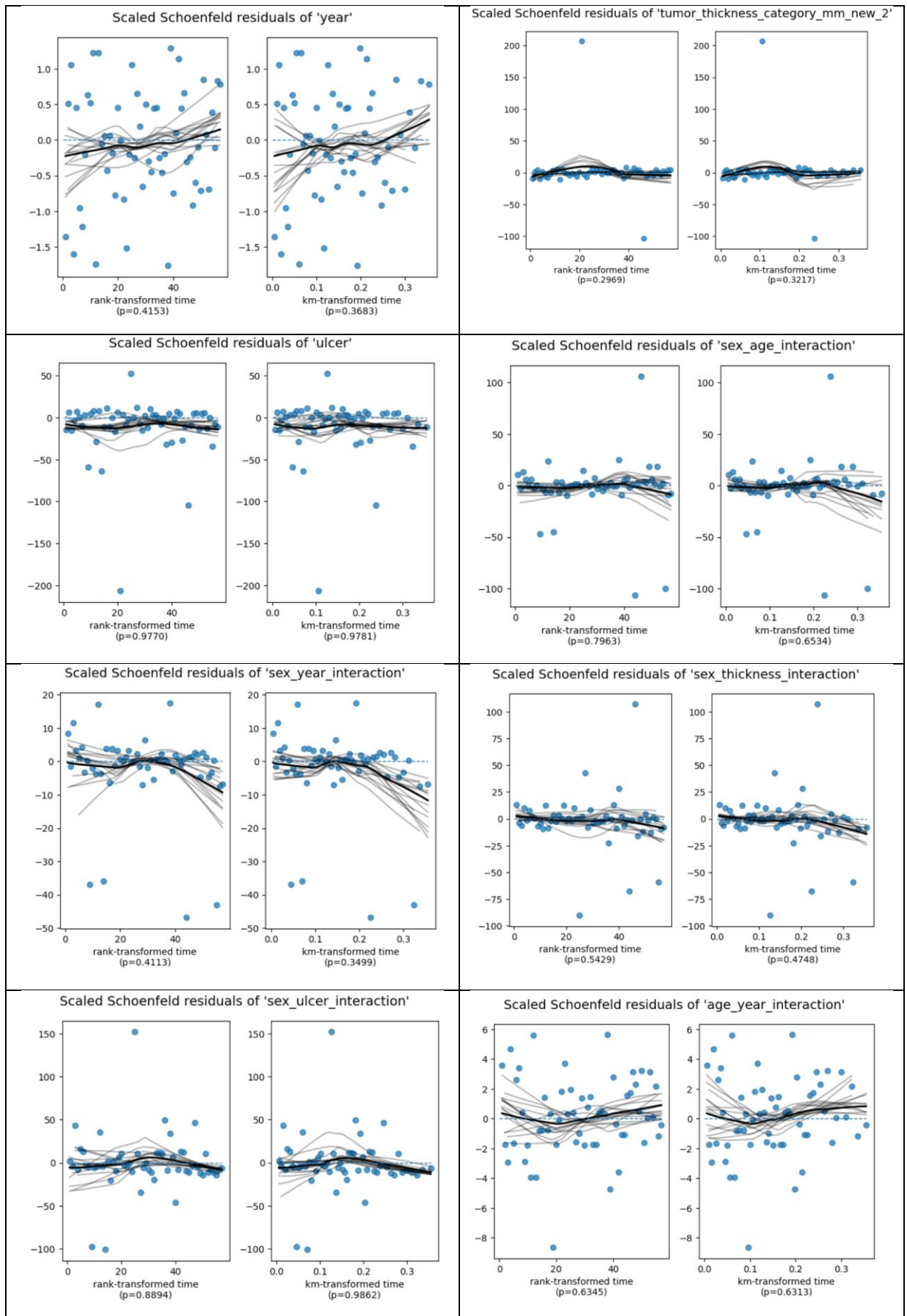
### 8.3.3.Assumption check binned model after elimination

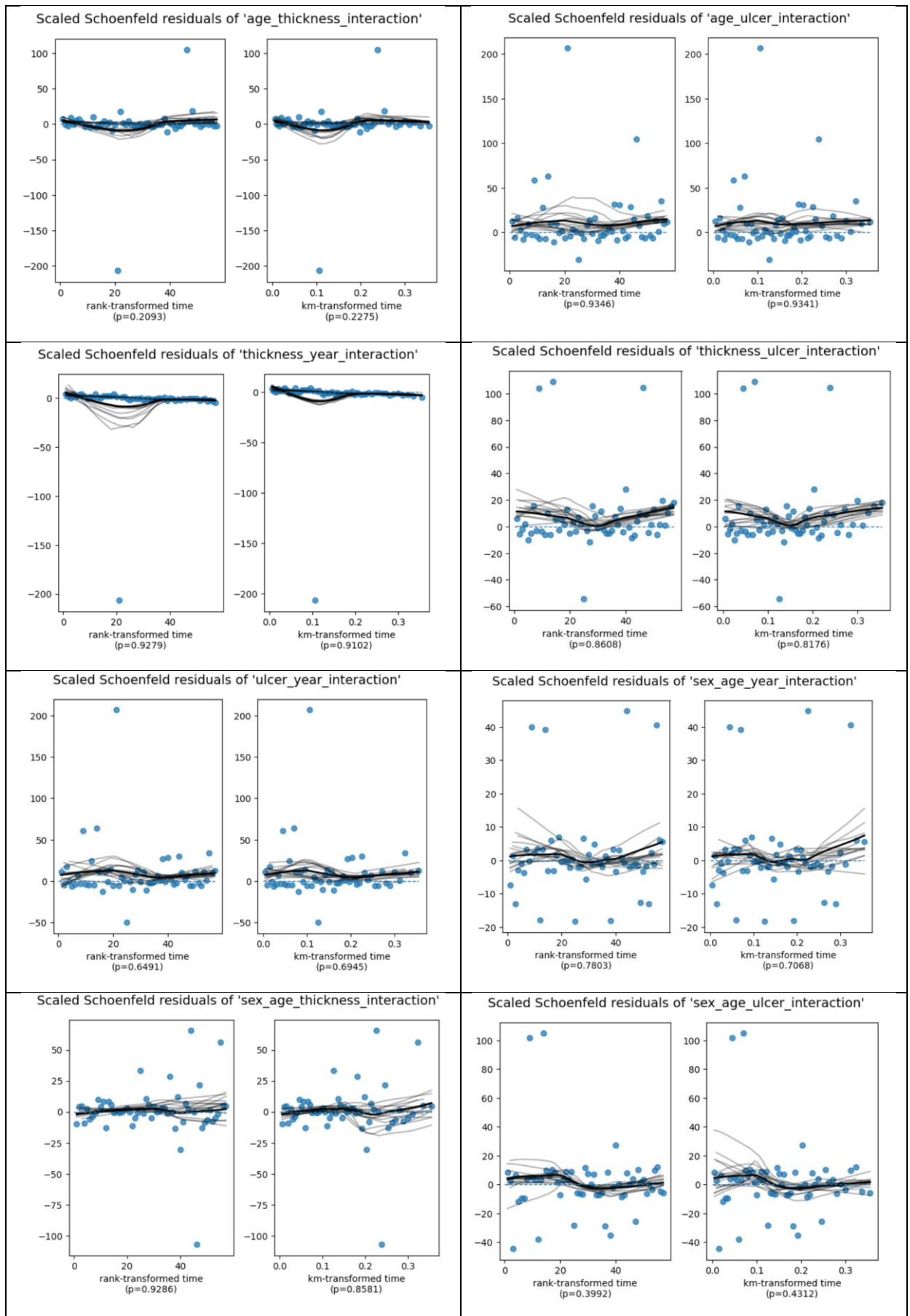


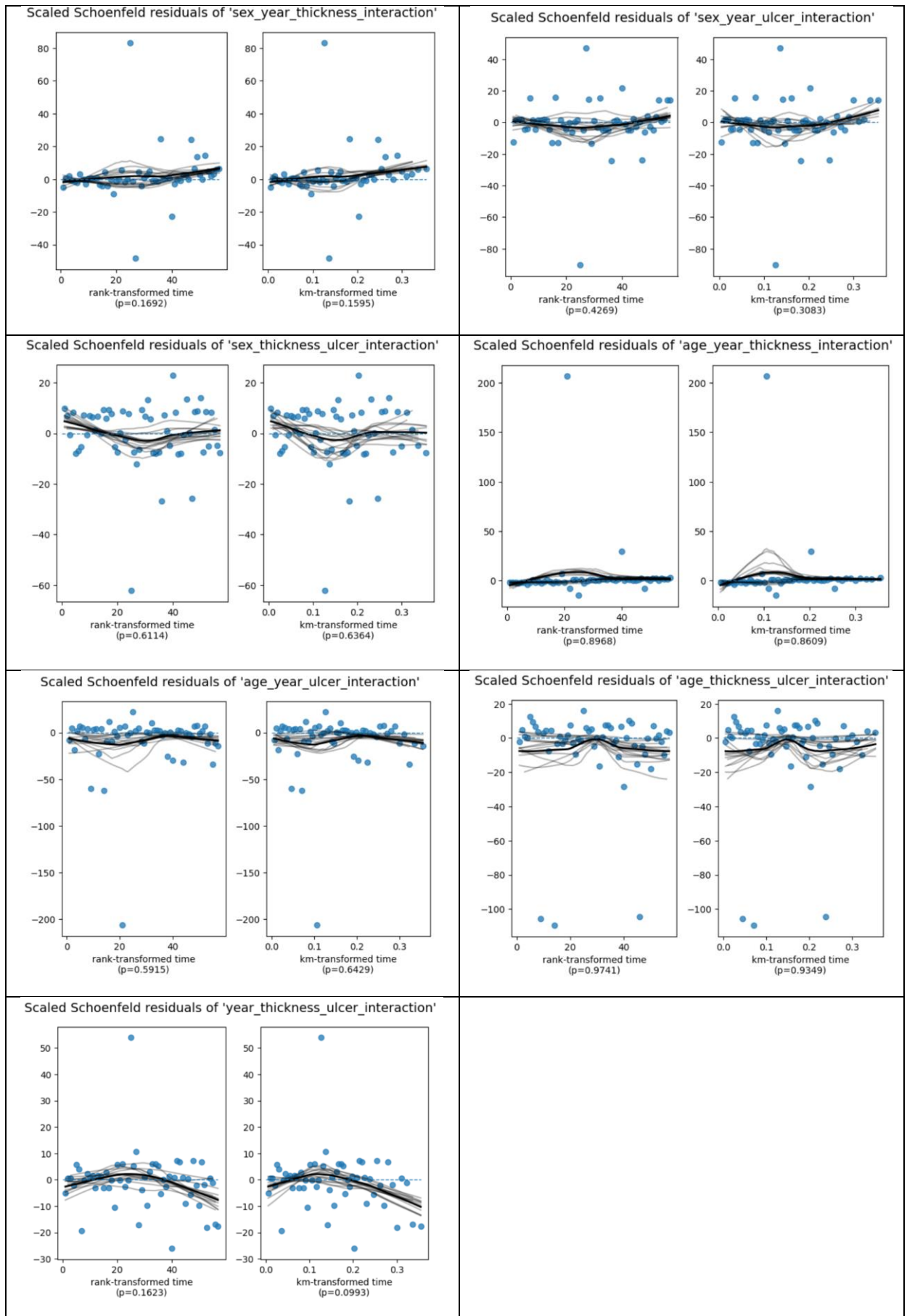
(Back to [Proportional Hazards Assumption Check Binned Model After Elimination](#))

### 8.3.4.Assumption check full interactions model





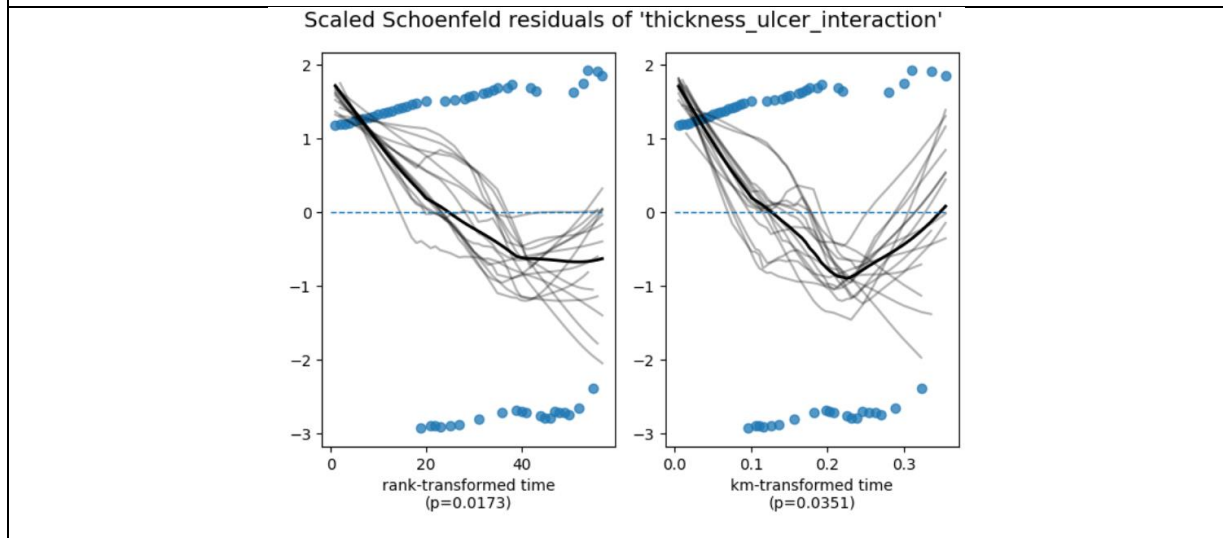




(Back to [Proportional Hazards Assumption Check Full Interactions Model](#))

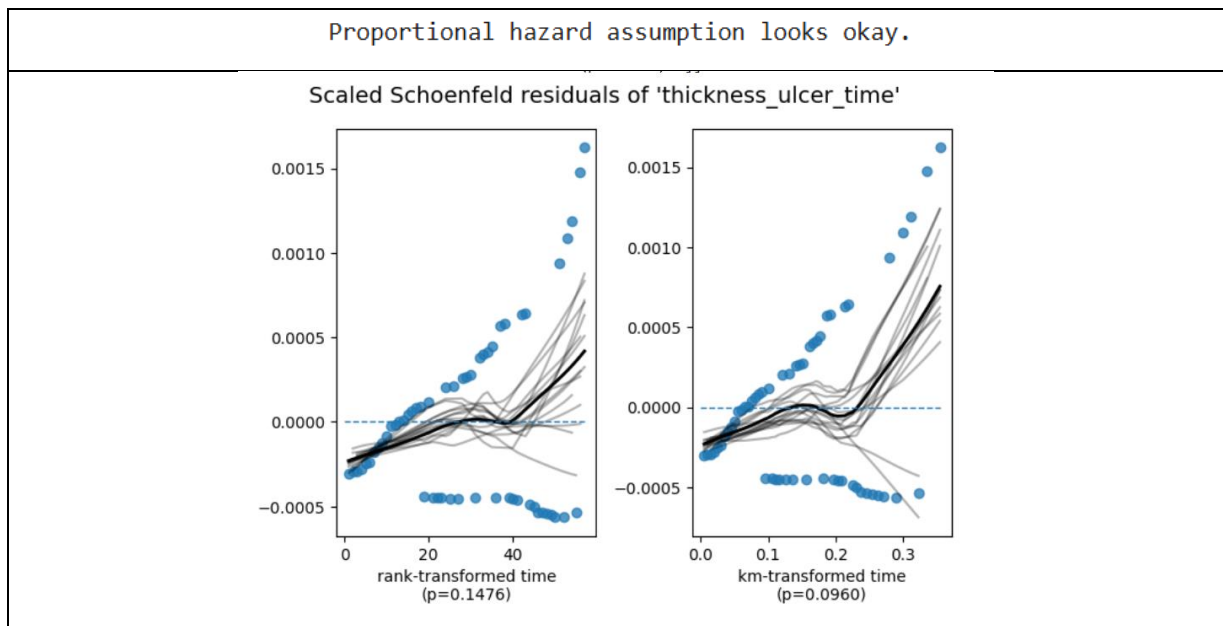
### 8.3.5.Assumption check full interaction model after elimination

1. Variable 'thickness\_ulcer\_interaction' failed the non-proportional test: p-value is 0.0173.



(Back to [Proportional Hazards Assumption Check Full Interaction Model After Elimination](#))

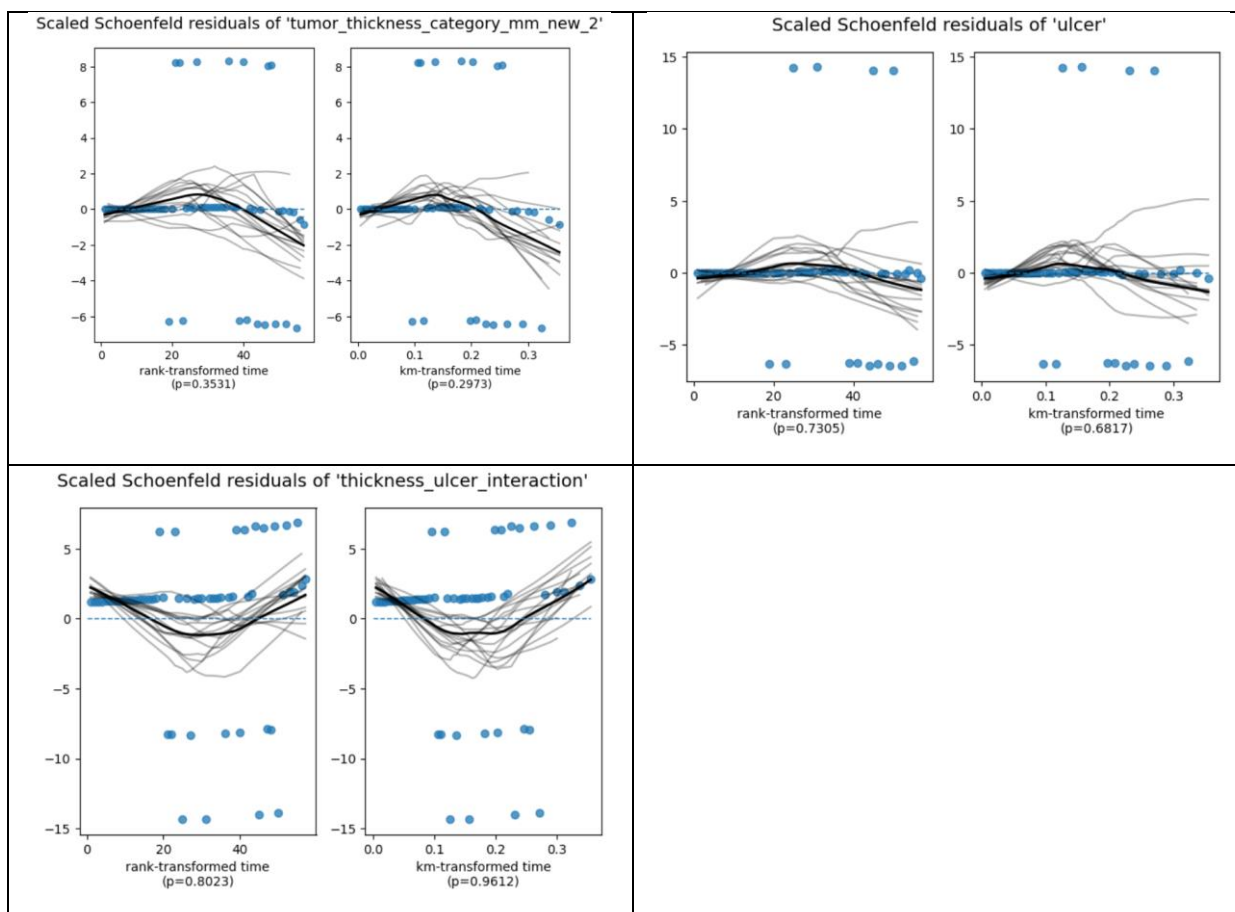
### 8.3.6.Assumption check full interaction model after elimination and adjusting the variable over time



(Back to [Proportional Hazards Assumption Check Full Interaction Model After Elimination And Adjusting The Variable Over Time](#))

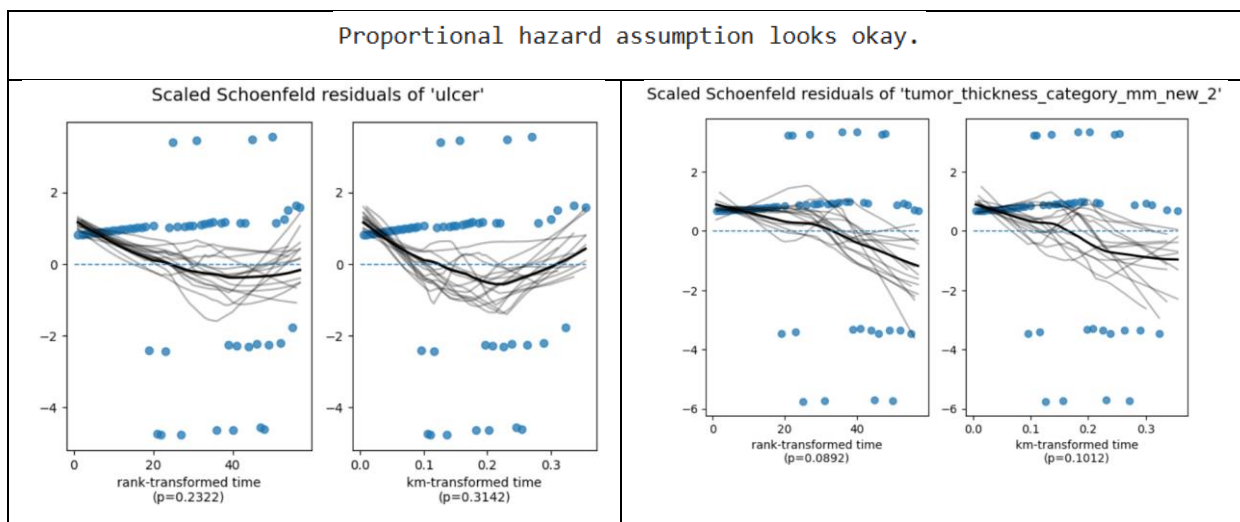
### 8.3.7.Assumption check partial interactions model

Proportional hazard assumption looks okay.



(Back to [Proportional Hazards Assumption Partial Interactions Model](#))

### 8.3.8.Assumption check partial interaction model after elimination



(Back to [Proportional Hazards Assumption Check Partial Interaction Model After Elimination](#))